

Contextuality as Structural Phase Dependence

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Abstract

We derive quantum contextuality as a necessary consequence of structural dependence on admissible phase geometry, rather than as a peculiarity of measurement choice or observer intervention. Within Phase Theory — the framework in which phase is the sole fundamental constituent of physical reality and all standard physical structures are emergent from global admissibility enforcement on a phase-bearing substrate — outcome assignments depend on the global admissibility context in which a distinction

is imposed, not on pre-existing local properties of an underlying system. We show that contextuality arises whenever admissible phase configurations admit multiple incompatible partitions of the admissible phase manifold $\mathcal{M}_{\text{adm}}$, and we recover all standard contextuality results — including Kochen–Specker-type impossibility theorems, state-independent contextuality, and Bell inequalities — without invoking wavefunctions, hidden variables, or measurement-induced collapse. Contextuality is thus revealed to be an objective structural feature of phase-consistent reality, not a signature of epistemic limitation or experimental design. We further derive the Peres–Mermin magic square, graph-theoretic contextuality inequalities, sheaf-cohomological obstructions to global outcome assignments, and contextuality monotones entirely within the Phase Theory framework. The first Betti number of $\mathcal{M}_{\text{adm}}$ is shown to be the topological invariant governing state-independent contextuality; the first Čech cohomology class of the outcome sheaf is the obstruction to Kochen–Specker-type assignments; and the Lovász theta function of the exclusivity graph is recovered as an admissibility invariant. Eight classes of falsifiable experimental predictions beyond standard quantum field theory and general relativity are enumerated, including contextuality reduction across symmetry-breaking phase transitions, dark-sector contextuality imprints in recoil spectra, and contextuality scaling with entanglement entropy in many-body systems.

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1. Why Contextuality Must Be Structural

The phenomenon of contextuality in quantum mechanics has resisted satisfactory explanation since its formal identification in the work of Kochen and Specker [1]. In the standard operational picture, contextuality is characterized as the dependence of measurement outcomes on the co-measured set of compatible observables — as though the result of ascertaining the value of an observable quantity were sensitive to which other quantities are examined simultaneously. This formulation, while technically precise, leaves unaddressed the deeper question: what is it about physical systems that necessitates such dependence? Is contextuality a feature of the act of observation, of the epistemic limitations of observers, or of something intrinsic to the constitution of reality itself?

The standard interpretation frameworks have uniformly failed to provide an ontologically satisfying answer to this question. Copenhagen quantum mechanics treats contextuality as a consequence of the indivisibility of measurement from the system observed, providing no further structural account [21]. Hidden variable programs — most notably Bohmian mechanics — reproduce quantum statistics but do so by relegating contextuality to the quantum force acting on hidden particle positions, thereby importing contextuality as an empirical datum rather than deriving it from fundamental principles [22]. Operational approaches, such as that of Spekkens [6], provide a mathematically rigorous characterization of contextuality in terms of preparations, transformations, and measurements, but explicitly decline to

furnish an ontological account, treating contextuality as a feature of empirical models rather than of physical structure.

This paper adopts a fundamentally different stance. We argue that the persistence of the explanatory deficit surrounding contextuality is traceable to a structural error at the level of ontological presupposition: the assumption that physical systems possess, prior to any contextual determination, definite values of observable quantities. Once this assumption — which we term *value realism* — is relinquished in favor of a global structural ontology, contextuality ceases to be a puzzling and apparently paradoxical feature of quantum systems and becomes, instead, a theorem derivable from first principles.

Phase Theory [11], the framework developed in this series, provides precisely such a global structural ontology. In Phase Theory, the sole fundamental constituent of physical reality is phase. All structures that appear as objects in standard physics — particles, fields, spacetime geometry, gauge symmetries, the particle spectrum — are emergent from the global organization, topology, and mutual consistency of a phase-bearing substrate. The governing principle is global consistency enforcement: physical laws arise as effective descriptions valid in regimes where coherent propagating phase modes dominate, and all apparent features of quantum mechanics — including superposition, interference, entanglement, and probability — arise from admissibility constraints on this substrate.

Within this framework, the question of contextuality acquires a precise and answerable form. A context is not an experimental procedure imposed from outside a system; it is a specific admissible partition of phase configuration space, itself a structural object of the theory. The dependence of outcomes on context is therefore not a dependence on anything external to physical reality — it is a dependence on the global admissibility structure of reality itself. Contextuality is not a feature added to Phase Theory to account for quantum

phenomena; it is a theorem derived from the topology of the admissible phase manifold.

The central thesis of this paper is the following: *contextuality is a necessary structural consequence of global admissibility enforcement on a phase-bearing substrate, and all known contextuality results in quantum mechanics are special cases of this single foundational fact.* The apparent strangeness of contextuality — its resistance to classical intuition, its apparent dependence on observer choices, its non-classical character — dissolves once the appropriate structural ontology is in place. A universe constituted by phase, governed by global admissibility, cannot but exhibit contextuality. The question is not why quantum systems are contextual, but why any description of phase-consistent reality that ignores global admissibility structure would expect systems to be noncontextual in the first place.

We distinguish throughout between *operational contextuality* — the dependence of outcome statistics on experimental co-measurement context, as studied in the tradition of Spekkens [6], Abramsky and Brandenburger [7], and Cabello, Severini and Winter [8] — and *structural contextuality*, the Phase Theory concept introduced in this paper. Structural contextuality is the more fundamental notion: it is the impossibility of a global assignment of outcomes consistent with all admissible partitions of phase configuration space. We show in Section 5 that structural contextuality implies and generalizes operational contextuality, and reduces to it in the empirical domain.

The paper is organized as follows. Sections 2 through 6 establish the mathematical foundations: the formal definition of contexts, the admissible phase manifold, and the topology governing incompatibility. Sections 7 through 14 derive all standard contextuality results as theorems of Phase Theory. Sections 15 through 24 develop the connections to experiment, computation, and information theory. Sections 25 through 33 extend the

framework to relativistic, many-body, and dark-sector regimes. Sections 34 through 38 enumerate falsifiable predictions, compare with alternative approaches, and provide the philosophical synthesis. A full bibliography of primary sources is provided at the close.

2. What "Context" Means in Phase Theory

The term "context" appears throughout the quantum foundations literature, but invariably as an operationally defined notion: a context is a set of compatible observables that are measured jointly in a given experimental run [1, 6]. This definition makes contexts artifacts of experimental design — objects that exist only because experimenters make choices. Phase Theory requires a structural definition of context that eliminates any dependence on agents or experimental procedures. We provide such a definition here and show that it recovers the operational notion as a limiting case.

Definition 33.1 (Context)

Let C denote the phase configuration space and $M_{\text{adm}} \subset C$ the admissible phase manifold (defined formally in Section 3). A **context** is a globally admissible partition P of M_{adm} induced by a compatible set of distinctions. Formally, a context is a pair (P, D) where: (i) $D = \{D_1, \dots, D_k\}$ is a set of distinction operators on M_{adm} ; (ii) each D_i is an admissibility-preserving map $D_i: M_{\text{adm}} \rightarrow \{0,1\}$; (iii) the partition P of M_{adm} is induced by the joint level sets of all D_i ; and (iv) the joint partition is globally admissible — i.e., each cell of P is a connected admissible region of M_{adm} .

This definition makes contexts structural objects. A context specifies three things simultaneously: (i) how phase space is carved into equivalence classes

by the given distinctions; (ii) which equivalence classes are distinguished from one another; and (iii) which distinctions are jointly admissible — i.e., which distinctions can be simultaneously imposed without violating global consistency constraints. None of these three specifications makes reference to an agent, observer, or experimental choice. Contexts are part of the structure of $\mathcal{M}_{\text{adm}}$, not additions to it.

A *distinction operator* $D: \mathcal{M}_{\text{adm}} \rightarrow \{0,1\}$ partitions $\mathcal{M}_{\text{adm}}$ into two globally admissible regions: $D^1(0)$ and $D^1(1)$. The structural analog of a quantum projection operator is a distinction operator: just as a projector P on a Hilbert space partitions the space into eigenspaces corresponding to eigenvalues 0 and 1, a distinction operator partitions $\mathcal{M}_{\text{adm}}$ into two phase-consistent regions. We show in Section 10 that the correspondence is exact: every projection operator on a Hilbert space of dimension ≥ 3 corresponds to a well-defined distinction operator on the appropriate $\mathcal{M}_{\text{adm}}$.

The set of all contexts forms a partially ordered set (poset) under refinement. Context (P_1, D_1) refines context (P_2, D_2) if P_1 is finer than P_2 — i.e., every cell of P_1 is a subset of some cell of P_2 . The partial order structure of contexts is denoted $\text{Ctx}(\mathcal{M}_{\text{adm}})$. This poset is the correct generalization of the lattice of projection operators in quantum mechanics to the Phase Theory setting. The maximal contexts in $\text{Ctx}(\mathcal{M}_{\text{adm}})$ correspond to maximal compatible sets of distinctions — the analog of complete orthonormal bases in Hilbert space.

Definition 33.2 (Context Compatibility)

Two contexts (P_1, D_1) and (P_2, D_2) are **compatible** if there exists a context $(P_{12}, D_1 \cup D_2)$ that is a common refinement of both — i.e., P_{12} is globally admissible and is finer than both P_1 and P_2 . They are **incompatible** if no such admissible common refinement exists.

Context incompatibility is the Phase Theory analog of non-commutativity in quantum mechanics. Two contexts are compatible if and only if their induced partitions can be simultaneously imposed on M_{adm} without violating global admissibility. When they are incompatible, any attempt to simultaneously impose both partitions on M_{adm} necessarily produces regions that violate global consistency constraints — topological barriers to joint assignment. The incompatibility of contexts is an objective structural fact about M_{adm} , not a limitation of any observer's experimental apparatus.

We provide a worked example in the two-dimensional case. Consider a phase configuration space $C = \mathbb{R}^2$ with the admissibility constraint that configurations must lie on the unit circle: $M_{\text{adm}} = S^1$. Two natural distinction operators on S^1 are $D_x: \varphi \mapsto \theta(\cos \varphi)$ (partitioning by the sign of the x-component) and $D_y: \varphi \mapsto \theta(\sin \varphi)$ (partitioning by the sign of the y-component). Both are individually admissible on S^1 . Their joint partition into four quadrants is also admissible — S^1 is divided into four connected arcs. Hence D_x and D_y are compatible in this example. The three-dimensional case — with $M_{\text{adm}} = S^2$ — provides the first example of incompatible distinctions, corresponding to non-commuting spin observables, as developed in Section 10.

In the quantum mechanical limit, distinction operators correspond to projection operators, contexts correspond to maximal commuting sets of observables, and context compatibility corresponds to commutativity. The Phase Theory formalism is therefore a strict generalization of the quantum mechanical framework, in which the algebraic structure of observables is derived from the topological structure of M_{adm} rather than postulated. This derivation is completed in Section 10.

We close this section by noting the relationship to the measurement algebra of quantum mechanics. The measurement algebra of a quantum system is the

algebra of bounded self-adjoint operators on the system's Hilbert space, organized by the commutative subalgebra structure. Each maximal commutative subalgebra corresponds to a maximal context in $\text{Ctx}(\mathcal{M}_{\text{adm}})$. The partition lattice of $\mathcal{M}_{\text{adm}}$ is the Phase Theory object that generates the measurement algebra structure in the quantum limit. This provides a derivation of the algebraic structure of quantum mechanics from phase topology — a key result of the Phase Theory program.

3. The Admissible Phase Manifold: Formal Setup

The admissible phase manifold $\mathcal{M}_{\text{adm}}$ is the mathematical heart of Phase Theory. All physical content of the theory — particle spectrum, interaction structure, quantum behavior, contextuality, gravity — is encoded in the geometry and topology of $\mathcal{M}_{\text{adm}}$. In this section we provide the formal mathematical definition and establish the key topological properties that will underwrite the contextuality results of subsequent sections.

Let $\mathcal{C}$ denote the full phase configuration space. $\mathcal{C}$ is a smooth infinite-dimensional manifold whose points are phase configurations — assignments of a phase value $\varphi(x) \in S^1$ to each point in the underlying topological substrate. The substrate itself is not presupposed to have the structure of space or time; these structures emerge from the global organization of phase modes, as detailed in Papers 1-5 of this series [11]. The full configuration space $\mathcal{C}$ carries a natural topology induced by the L^2 norm on phase differences.

Definition 33.3 (Phase Consistency Functional)

The **phase consistency functional** $\Phi: \mathcal{C} \rightarrow \mathbb{R}$ is a functional on phase configuration space satisfying: (i) Φ is smooth and bounded below; (ii) $\Phi[c] =$

0 if and only if the configuration c satisfies all global consistency constraints of Phase Theory; (iii) $\Phi[c] > 0$ for all inconsistent configurations; (iv) Φ is covariant under the symmetry group of Phase Theory. The **admissible phase manifold** is defined as $M_{\text{adm}} = \{c \in C : \Phi[c] = 0\}$.

The phase consistency functional Φ encodes all global constraints of Phase Theory: phase coherence requirements, topological winding number conservation, boundary conditions on the substrate, and consistency of emergent structures. The zero-level set of Φ defines M_{adm} as the submanifold of physically realizable phase configurations. Inadmissible configurations ($\Phi[c] > 0$) are not physically realized — they correspond to phase assignments that would require global inconsistencies, analogous to configurations that violate gauge constraints or topological conservation laws.

$$(3.1) \text{ [Equation: } M_{\text{adm}} = \Phi^{-1}(0) = \{c \in C : \Phi[c] = 0\}]$$

The topology of M_{adm} is generically non-trivial. We establish this by the following argument. The full configuration space C is contractible (being an infinite-dimensional vector space in the relevant topological model). However, the constraint $\Phi[c] = 0$ is generically not satisfiable by all configurations simultaneously — the level set of Φ at 0 cuts out a proper submanifold with non-trivial topology. The topology of M_{adm} is determined by the winding structure of global phase modes, which generically produces non-contractible loops, non-trivial higher homotopy groups, and stratified boundary structure.

Proposition 33.1 (Topology of M_{adm})

For any phase configuration space C with more than one globally consistent

winding sector, M_{adm} is a stratified manifold with non-trivial fundamental group $\pi_1(M_{\text{adm}}) \neq 1$. In particular, M_{adm} admits at least two non-homotopic loops, corresponding to distinct winding sectors of the phase substrate.

Proof. Let $c_0, c_1 \in M_{\text{adm}}$ be two phase configurations in distinct winding sectors — configurations that cannot be continuously deformed into one another while maintaining $\Phi[c] = 0$ throughout the deformation. Such configurations exist by assumption (more than one winding sector). Any path $\gamma: [0,1] \rightarrow C$ connecting c_0 to c_1 must pass through a configuration with $\Phi[c] > 0$ — i.e., must exit M_{adm} . Hence c_0 and c_1 lie in different connected components of M_{adm} , or — in the connected case — are connected by paths that generate non-trivial elements of π_1 . In either case, $\pi_1(M_{\text{adm}}) \neq 1$. $\square$

The stratification of M_{adm} is the decomposition of M_{adm} into strata of decreasing dimension, separated by admissibility boundaries. The top stratum consists of phase configurations satisfying all constraints with strict inequalities (interior of admissibility). The boundary strata consist of configurations at the boundary of admissibility — phase configurations that satisfy the constraints exactly but lie at the edge of the admissible region, where small perturbations would render them inadmissible. The structure of these strata governs the coarse-grained particle spectrum of Phase Theory: each stratum corresponds to a class of phase modes with distinct topological properties [11].

The partition lattice $L(M_{\text{adm}})$ of M_{adm} is the lattice of all admissible partitions of M_{adm} , ordered by refinement. This lattice governs the structure of admissible contexts: every context in $\text{Ctx}(M_{\text{adm}})$ corresponds to an element of $L(M_{\text{adm}})$, and context compatibility corresponds to the existence of a common refinement in $L(M_{\text{adm}})$.

Theorem 33.0 (Partition Lattice and Contextuality Precondition)

If $\pi_1(M_{\text{adm}})$ is non-trivial, then $L(M_{\text{adm}})$ admits incompatible maximal elements — i.e., there exist maximal admissible partitions $P_1, P_2 \in L(M_{\text{adm}})$ with no common admissible refinement. This is the structural precondition for contextuality.

Proof. Non-trivial $\pi_1(M_{\text{adm}})$ implies the existence of a non-contractible loop γ in M_{adm} . Two maximal partitions P_1 and P_2 are constructed by taking complementary coarsenings of the partition induced by the winding number along γ . Any common admissible refinement of P_1 and P_2 would require a partition that simultaneously respects both sets of cells. But since γ is non-contractible, any such joint partition would need to assign consistent winding labels globally — which contradicts the non-triviality of the loop. Hence no admissible common refinement exists. $\square$

This theorem establishes the structural precondition for contextuality at the level of the admissible phase manifold itself. The non-trivial topology of M_{adm} is not an accident of particular quantum systems — it is a generic feature of any phase-bearing substrate with more than one consistent winding sector. The precondition for contextuality is therefore satisfied universally in Phase Theory, and contextuality is a universal structural feature of phase-consistent reality.

4. Structural Incompatibility of Partitions

We now develop the formal theory of partition incompatibility on M_{adm} . The key insight is that incompatibility — the structural precondition for contextuality — is a topological property of M_{adm} , not an operational property of any experimental procedure. Incompatibility exists in the structure of

phase-consistent reality independently of whether any agent attempts to resolve it through measurement.

Definition 33.4 (Compatible and Incompatible Partitions)

Two admissible partitions $P_1, P_2 \in \mathcal{L}(\mathcal{M}_{\text{adm}})$ are **compatible** if there exists an admissible partition $P_{12} \in \mathcal{L}(\mathcal{M}_{\text{adm}})$ such that P_{12} refines both P_1 and P_2 — i.e., each cell of P_{12} is a subset of a cell of P_1 and a subset of a cell of P_2 . They are **incompatible** if no such admissible common refinement exists.

Theorem 33.1 (Incompatibility Theorem)

Two admissible partitions P_1, P_2 of $\mathcal{M}_{\text{adm}}$ are incompatible if and only if the intersection of their induced equivalence classes does not define a globally admissible partition — i.e., if and only if there exist cells $A \in P_1$ and $B \in P_2$ such that $A \cap B$ is non-empty but not admissible (it intersects an admissibility boundary).

Proof. ($\Rightarrow$) Suppose P_1 and P_2 are incompatible. Then for any proposed refinement P_{12} , there exists a cell $A \times B$ of the product partition (where $A \in P_1, B \in P_2$) that is non-empty but whose closure intersects an admissibility boundary of $\mathcal{M}_{\text{adm}}$. This intersection is therefore not in the interior of $\mathcal{M}_{\text{adm}}$ and cannot serve as a cell of an admissible partition. Hence P_{12} is not admissible. ($\Leftarrow$) Suppose the intersection of induced equivalence classes does define globally admissible cells. Then the partition into cells $A \cap B$ (for all $A \in P_1, B \in P_2$ with $A \cap B$ non-empty) is a well-defined admissible common refinement of P_1 and P_2 . This contradicts incompatibility. $\square$

Theorem 33.1 establishes that partition incompatibility is a topological property of M_{adm} . The condition " $A \cap B$ intersects an admissibility boundary" is a condition on the geometry of M_{adm} , not on any experimental procedure. This is the formal content of the claim that contextuality is structural.

To illustrate, consider a two-qubit analogue in Phase Theory. The admissible phase manifold for a two-qubit-analogue system is topologically equivalent to S^3 (the three-sphere) after accounting for the global consistency constraints. The partitions induced by the three Pauli-analogue distinction operators $\sigma_x, \sigma_y, \sigma_z$ on S^3 have the following compatibility structure: any two are compatible (their joint partition is admissible), but no three distinct pairs of Pauli distinctions across two subsystems yield a globally admissible four-way partition. This is the Phase Theory origin of the failure of joint probability distributions for non-commuting observables.

The incompatibility relation on the set of maximal partitions of M_{adm} defines a graph G_{incomp} : vertices are maximal admissible partitions, and an edge connects two vertices if and only if the corresponding partitions are incompatible. The chromatic number $\chi(G_{\text{incomp}})$ measures the minimum number of colors needed to color the vertices so that incompatible partitions receive different colors. We identify $\chi(G_{\text{incomp}})$ as the *contextual complexity* of the system.

$$(4.1) \text{ [Equation: } \chi(G_{\text{incomp}}) = \min\{k : \exists \text{ k-coloring of maximal partitions of } M_{\text{adm}} \text{ consistent with compatibility}\}]$$

The contextual complexity $\chi(G_{\text{incomp}})$ is an invariant of M_{adm} : it does not depend on any choice of basis, experimental setup, or reference frame. In the classical limit (Section 14), $\chi(G_{\text{incomp}}) = 1$ — all partitions are compatible and a single color suffices. In the quantum limit, $\chi(G_{\text{incomp}}) = 2$ for single qubits and increases with system dimension. This provides a precise quantitative

measure of contextuality in Phase Theory, reducing to the known graph-theoretic measures in the quantum limit [8].

The incompatibility graph G_{incomp} also encodes the structure of the Kochen-Specker theorem: the KS theorem is equivalent to the statement that G_{incomp} cannot be two-colored in a manner consistent with the outcome constraints — which is equivalent to the non-existence of a global noncontextual assignment. This connection is developed in Section 7. The graph-theoretic formulation of contextuality (CSW framework [8]) is recovered as a coarsening of the full Phase Theory partition lattice structure, as shown in Section 18.

5. Definition of Contextuality (Phase-Theoretic)

We are now in position to state the central definition of the paper — the Phase Theory definition of structural contextuality. This definition makes no reference to agents, measurements, experimental procedures, or epistemic states. It is a purely structural characterization of a property of the admissible phase manifold.

Definition 33.5 (Structural Contextuality)

A system governed by admissible phase manifold M_{adm} is **structurally contextual** if there exists no single global assignment of outcomes $f: M_{\text{adm}} \times \text{Ctx}(M_{\text{adm}}) \rightarrow O$ (where O is the outcome set) such that: (i) for each context $C \in \text{Ctx}(M_{\text{adm}})$, the assignment $f(\cdot, C)$ is consistent with the admissible outcome set $O(C)$; and (ii) $f(\cdot, C)$ is independent of C — i.e., outcomes depend only on the phase configuration, not on the context in which it is distinguished.

Condition (ii) is the noncontextuality condition: outcomes depend only on the phase configuration and not on which context is used to distinguish it. Structural contextuality is the violation of condition (ii) — the necessary dependence of outcomes on context, arising from the impossibility of a context-independent global assignment.

The contrast with the operational definition (Spekkens [6]) is important and bears elaboration. Spekkens defines a model as noncontextual if the outcome of any measurement procedure depends only on the equivalence class of the procedure under operational equivalence. This definition is agent-relative: it depends on what it means for two procedures to be operationally equivalent, which requires reference to the procedures available to an experimenter. Definition 33.5 makes no such reference: it depends only on the structure of M_{adm} and the outcome sheaf over $\text{Ctx}(M_{\text{adm}})$.

Proposition 33.2 (Structural Implies Operational Contextuality)

Let a system be structurally contextual in the sense of Definition 33.5. Then in any empirical realization of the system — any regime where phase configurations map to quantum states and admissible partitions map to quantum observables — the system exhibits operational contextuality in the sense of Spekkens [6].

Proof. The empirical realization map sends each context $C \in \text{Ctx}(M_{\text{adm}})$ to a maximal commuting set of quantum observables, and each admissible partition to the joint spectral decomposition of those observables. Structural contextuality implies no global section of the outcome sheaf (shown below). Absence of a global section over $\text{Ctx}(M_{\text{adm}})$ implies, in the image of the realization map, no global noncontextual hidden variable model — which is precisely Spekkens' operational contextuality. $\square$

The converse of Proposition 33.2 also holds in the class of systems describable by Phase Theory: every operationally contextual system has a Phase Theory description with a structurally contextual admissible manifold. This establishes a full equivalence between structural and operational contextuality in the Phase Theory framework, with the advantage that the structural formulation is ontologically transparent.

Proposition 33.3 (No-Global-Assignment Theorem)

A system is structurally contextual if and only if the outcome sheaf F over $\text{Ctx}(M_{\text{adm}})$ admits no global section — i.e., there is no element of $F(\text{Ctx}(M_{\text{adm}}))$ that restricts to the correct outcome on every context.

Proof. By definition, a global section of F is exactly a global noncontextual assignment: a map that assigns to each context its admissible outcomes consistently across all contexts. The existence of such a section is equivalent to noncontextuality. Hence its non-existence is equivalent to structural contextuality. The sheaf formulation is developed in full in Sections 21–22. $\square$

Proposition 33.3 establishes the sheaf-cohomological formulation of contextuality as a corollary of Definition 33.5. The absence of a global section of the outcome sheaf is the Phase Theory version of the Kochen–Specker theorem, of Bell's theorem, and of all known impossibility results in quantum foundations. These are not distinct theorems requiring separate proofs — they are all instances of Proposition 33.3 applied to different admissible phase manifolds.

6. Phase Configuration Space: Global Topology and Stratification

The topological structure of phase configuration space is the fundamental arena in which all Phase Theory results are established. We provide here a detailed account of this topology, the stratification of the admissible submanifold, and the holonomy structure that encodes contextual information. This section provides the topological infrastructure for the contextuality results of Sections 7-14.

The full phase configuration space $\mathcal{C}$ is modeled as the space of smooth maps from the phase substrate X to the circle group S^1 : $\mathcal{C} = \text{Map}(X, S^1)$. Here X is the underlying topological substrate — not assumed to have the structure of spacetime, as spacetime itself emerges from phase organization in Phase Theory. The topology of $\mathcal{C}$ is the compact-open topology, making $\text{Map}(X, S^1)$ a topological group under pointwise multiplication of phases.

$$(6.1) \text{ [Equation: } \mathcal{C} = \text{Map}(X, S^1) \cong \{\varphi: X \rightarrow [0, 2\pi) : \varphi \text{ smooth}\}]$$

The connected components of $\mathcal{C}$ are indexed by the homotopy classes $[X, S^1] \cong H^1(X; \mathbb{Z})$ — the first cohomology of the substrate with integer coefficients. Each connected component corresponds to a distinct winding sector of the phase field. The admissibility constraint $\Phi[c] = 0$ does not distinguish between configurations in the same winding sector, but it does impose constraints within each sector.

The stratification of $\mathcal{M}_{\text{adm}}$ proceeds as follows. Let Σ_0 denote the top stratum — phase configurations in the interior of the admissible region, far from any admissibility boundary. Let Σ_1 denote the codimension-1 boundary stratum — configurations at a single admissibility boundary. Let Σ_k denote the codimension- k stratum, the intersection of exactly k admissibility boundaries. The stratification is:

$$(6.2) \text{ [Equation: } \mathcal{M}_{\text{adm}} = \Sigma_0 \cup \Sigma_1 \cup \Sigma_2 \cup \cdots \cup \Sigma_N]$$

Each stratum Σ_k is a smooth submanifold of $\mathcal{C}$ of decreasing dimension. The top stratum Σ_0 is open and dense in M_{adm} . The boundary strata are closed in Σ_0 and carry information about the particle spectrum of Phase Theory: the codimension-1 boundary strata correspond to the marginally stable phase modes identified as particle species in Paper 11 of this series [11].

Theorem 33.A (Non-Abelian Fundamental Group)

For systems with more than two pairwise incompatible admissible partitions, the fundamental group $\pi_1(M_{\text{adm}})$ is non-abelian.

Proof. Each incompatible partition pair contributes a generator to $\pi_1(M_{\text{adm}})$, corresponding to a loop that traverses the admissibility boundary between the two partition sectors. For three or more pairwise incompatible partitions, the generators g_{12} , g_{13} , g_{23} satisfy a relation of the form $g_{12} \circ g_{23} = g_{13} \circ \delta$ where δ is a non-trivial central element. This prevents $\pi_1(M_{\text{adm}})$ from being abelian, since the generators do not commute. For the case of three Pauli-analogue partitions on the spin-1 admissible manifold, the fundamental group is the quaternion group Q_8 , which is non-abelian. $\square$

The holonomy structure of M_{adm} provides the most direct topological measure of contextuality. The admissibility connection ∇ on M_{adm} is a connection on the principal bundle of partition frames over M_{adm} . As a phase configuration traverses a loop γ in M_{adm} , the holonomy $\text{Hol}(\gamma)$ records the accumulated contextual transformation — the extent to which the admissible partitions have rotated relative to one another as the configuration varies.

$$(6.3) \text{ [Equation: } \text{Hol}(\gamma) = \text{P exp}(\oint_{\gamma} A_{\nabla}) \in \text{Aut}(\text{P}) \text{ for loop } \gamma \in \pi_1(M_{\text{adm}})]$$

The holonomy $\text{Hol}(\gamma)$ is trivial (the identity) if and only if γ is contractible in M_{adm} — i.e., if and only if the system is locally noncontextual along γ . For non-

contractible loops, $\text{Hol}(\gamma)$ is a non-trivial automorphism of the partition frame, corresponding to a contextual transformation. The holonomy group $\text{Hol}(\mathcal{M}_{\text{adm}})$ — the group generated by all such holonomies — is therefore a measure of the total contextual structure of the system. A system is noncontextual if and only if $\text{Hol}(\mathcal{M}_{\text{adm}})$ is trivial. This establishes the holonomy-contextuality correspondence: contextuality is precisely the non-triviality of the holonomy group of the admissibility connection.

7. Kochen-Specker Theorem Reinterpreted

The Kochen-Specker (KS) theorem [1] is one of the most celebrated results in quantum foundations. It states that in a Hilbert space of dimension ≥ 3 , there is no consistent assignment of values 0 or 1 to all projection operators such that (i) exactly one operator in each orthonormal basis receives the value 1, and (ii) the assignment is independent of which basis (context) the operator appears in. This is a purely mathematical result, but its physical implications — that quantum mechanics cannot be supplemented by a noncontextual hidden variable model — are profound [23].

The original proof by Kochen and Specker used 117 vectors in $\mathbb{R}^3$ [1]; subsequent proofs by Peres [4], Mermin [5], Conway and Kochen [17], and others have reduced this number and clarified the mathematical structure. Yu and Oh [10] provided a state-independent proof using 13 rays. Despite this progress, the theorem remained an empirical discovery about quantum mechanics rather than a derived consequence of any deeper principle — until now.

Proposition 33.4 (KS from Phase Theory)

If M_{adm} admits incompatible admissible partitions, then no global noncontextual outcome assignment exists. In particular, in any Phase Theory system whose admissible phase manifold corresponds to a Hilbert space of dimension ≥ 3 in the quantum limit, no global 0/1 assignment consistent with all admissible partitions can be noncontextual.

Proof. Suppose a global noncontextual assignment $f: M_{\text{adm}} \rightarrow \{0,1\}$ exists, satisfying condition (ii) of Definition 33.5. Then f defines a global section s of the outcome sheaf F over $\text{Ctx}(M_{\text{adm}})$: for each context C , the restriction $f|_C$ provides the local section over C . By Proposition 33.3, the existence of a global section implies noncontextuality. But by the cohomological obstruction theorem (Section 21), a global section of F exists if and only if the obstruction class $[\sigma] \in H^1(\text{Ctx}(M_{\text{adm}}); F)$ vanishes. For M_{adm} corresponding to dimension ≥ 3 , $[\sigma] \neq 0$ (proven in Section 21). Hence no global section exists, and no global noncontextual assignment exists. $\square$

Corollary 33.1 (KS as Corollary of Phase Topology)

The Kochen–Specker theorem is a corollary of the topology of M_{adm} . It is not a free-standing impossibility result about quantum mechanics — it is the quantum-limit expression of the general impossibility of global noncontextual assignments on admissible phase manifolds with non-trivial fundamental groups.

This reinterpretation is philosophically significant. The KS theorem, as originally formulated, appears to be a peculiarity of quantum mechanics: it requires the full apparatus of Hilbert space theory, projection operators, and orthonormal bases to state, and its proof requires explicit construction of 117 (or more carefully, 13 [10]) specific vectors. In Phase Theory, the KS theorem

requires none of this apparatus. It is a special case of the topological fact that non-trivial $\pi_1(M_{\text{adm}})$ obstructs global sections of the outcome sheaf.

Remark 33.1

The dimension-3 threshold in the KS theorem corresponds, in Phase Theory, to the minimum topological complexity required for $\pi_1(M_{\text{adm}})$ to be non-abelian and for the obstruction class $[\sigma]$ to be non-zero in $\mathbb{Z}/2\mathbb{Z}$ cohomology. For dimension-2 systems (qubits), the admissible manifold is topologically simpler and the $\mathbb{Z}/2\mathbb{Z}$ obstruction class vanishes — consistent with the known fact that single-qubit systems admit noncontextual hidden variable models. This dimension dependence is thus explained topologically.

The reinterpretation of KS also removes what Mermin famously called "the peculiarity of quantum mechanics" [5]. There is nothing peculiar about quantum systems exhibiting KS contextuality. A universe governed by global admissibility of phase cannot be otherwise. The peculiarity belonged to the assumption of value realism — the assumption that quantum systems should be assigned definite pre-measurement values. Phase Theory shows this assumption is incompatible with the topology of any globally admissible phase-bearing substrate of the relevant complexity.

The Conway-Kochen Free Will Theorem [17] extends the KS result to entangled systems. Within Phase Theory, the Free Will Theorem is the statement that the contextual structure of the composite admissible manifold $M_{\text{adm}} \otimes M_{\text{adm}}$ is at least as rich as the contextual structure of either factor. This follows immediately from the tensor product structure of admissibility constraints (developed in Section 11). No special assumptions about free will are required — the "freedom" of experimental choice is replaced by the structural freedom in imposing admissible partitions.

8. No Pre-Existing Values

The Phase Theory response to the question of pre-existing values — the question of what is "really there" before contextual distinctions are imposed — is neither a return to hidden variables nor an endorsement of anti-realism. Phase Theory provides a positive ontology of outcomes that avoids both the Einstein-Podolsky-Rosen [24] realism trap and the Copenhagen refusal to engage with ontological questions.

The EPR argument [24] assumed that if the outcome of a measurement on a system can be predicted with certainty without disturbing the system, then there must be a pre-existing element of reality corresponding to that outcome. This assumption — which Einstein regarded as the criterion of physical reality — is exactly the assumption of value realism, which is what the KS theorem and Bell's theorem refute. Phase Theory dissolves this argument by replacing its ontological presupposition: outcomes are not revelations of pre-existing values but rather admissibility-consistent selections within a given contextual partition.

Definition 33.6 (Admissible Outcome Set)

For a context $C = (P, D) \in \text{Ctx}(\mathcal{M}_{\text{adm}})$, the **admissible outcome set** $\mathcal{O}(C)$ is the set of globally consistent phase configurations compatible with the partition P induced by C . Formally, $\mathcal{O}(C) = \{\text{cells of } P \text{ that are non-empty, connected, and globally admissible subsets of } \mathcal{M}_{\text{adm}}\}$. A **selection** is an element $s \in \mathcal{O}(C)$ — a specific admissible cell of the context partition to which the system's phase configuration belongs.

Selections are not revelations of hidden truths. There is no fact of the matter, prior to context imposition, about which cell of P the system occupies — because the cells of P do not exist as independent entities prior to the imposition of P . The partition is imposed structurally (by the admissibility

constraints of context C), and the selection is the consistent determination of which cell the current phase configuration falls within. This is a structural event, not a discovery.

Theorem 33.2 (No Pre-Existing Value Theorem)

In a structurally contextual system, there exists no assignment of pre-existing values to all admissible distinctions consistent with all admissible partitions.

Equivalently, for any phase configuration $c \in M_{\text{adm}}$, there is no function v_c :

$\text{Distinctions}(M_{\text{adm}}) \rightarrow \{0,1\}$ such that $v_c(D) = f(c, C)$ for all contexts C containing D , where f is the would-be noncontextual assignment.

Proof. If v_c existed for all $c \in M_{\text{adm}}$, then the map $f(c, C) = v_c(D)$ for any $D \in C$ would define a global noncontextual assignment. By Proposition 33.4 (and the KS theorem in the quantum limit), no such global noncontextual assignment exists for structurally contextual systems. Hence no v_c can exist. $\square$

The distinction between *selection* and *revelation* is central to the Phase Theory ontology. A revelation would be the uncovering of a pre-existing fact. A selection is the consistent determination of which admissible partition cell contains the current phase configuration — a determination that could not have been made before the partition was imposed, because the cells did not exist. This is not anti-realism: the phase configuration $c \in M_{\text{adm}}$ is real prior to context imposition. What is not real prior to context imposition is the cell label — because cell labels require cells, and cells require partitions, and partitions must be globally admissible.

The EPR argument fails precisely at this point. EPR argued that if a remote measurement can predict the outcome of a local measurement with certainty, then the local outcome must correspond to a pre-existing element of reality. But in Phase Theory, the remote measurement does not reveal a pre-existing

local value — it imposes a contextual partition on the composite admissible manifold $M_{\text{adm}} \otimes M_{\text{adm}}$, and the local selection is the consistent outcome within the induced local context. The correlation between remote and local outcomes is a correlation between selections, not between pre-existing values. This is entirely consistent with all Bell inequality violations, as we show in Section 11.

We further note that the distinction between outcome and property is essential here. A *property*, in the classical ontology, is a feature of an object that exists independently of whether or not that feature is ascertained. An *outcome*, in Phase Theory, is a context-relative selection from an admissible set. Properties, in the classical sense, do not exist in contextually structured phase systems. This is not a limitation of knowledge — it is an objective fact about the structure of admissible phase-consistent reality.

9. Measurement Without Privilege

One of the most persistent difficulties in quantum foundations is the privileged role assigned to measurement. In the standard quantum formalism, measurement is a primitive operation that cannot be reduced to anything more fundamental — it is the point at which the wave function "collapses," at which probabilistic potentiality is transmuted into definite actuality, and at which the Born rule assigns probabilities to outcomes. This privileged role generates the measurement problem [21, 25]: why does measurement produce definite outcomes when unitary evolution does not?

Phase Theory eliminates the privileged role of measurement entirely. In Phase Theory, a measurement is neither ontologically special nor epistemically privileged. It is a specific instance of context imposition — the structural process by which an admissible partition is applied to M_{adm} . The act of measuring is not the act of looking at reality from outside; it is the act of

imposing a structural partition on an admissible phase configuration. This imposition is itself a physical process governed by the same Phase Theory laws as everything else.

Theorem 33.3 (Measurement Neutrality)

In Phase Theory, the act of measurement is structurally equivalent to the imposition of a context — a globally admissible partition P of M_{adm} . It carries no ontological privilege beyond context-setting: no collapse occurs, no information is created by measurement that was not already encoded in the admissibility structure of M_{adm} , and no new physics enters at the moment of measurement.

Proof. By Definition 33.1, a context is a globally admissible partition of M_{adm} induced by a compatible set of distinctions. A measurement imposes exactly such a partition: it specifies which distinctions are simultaneously applied (the set of co-measured observables), and the outcome is the admissible selection within the induced cells. All information about the outcome — including the probabilities over possible selections — is encoded in the admissibility structure of M_{adm} prior to context imposition (see Section 13 for the probability law). Hence measurement adds no new structure not already present in M_{adm} , and carries no ontological privilege. $\square$

Corollary 33.2 (Contextuality Pre-exists Measurement)

Contextuality in a Phase Theory system pre-exists any measurement. The contextual structure is encoded in the topology of M_{adm} , not created by measurement acts. Measurement reveals contextual structure but does not produce it.

The measurement problem dissolves in Phase Theory because measurement is not special. Before measurement, the system occupies a phase configuration $c \in M_{\text{adm}}$ with a well-defined topological character. This configuration does not have a definite "value" for any distinction — not because it lacks definiteness, but because definite values of distinctions require contexts, and contexts are partitions of M_{adm} that have not yet been imposed. After measurement (context imposition), the system is associated with a specific cell of the partition — a context-relative definite outcome. No collapse is required because there is nothing that collapses: the transition from context-indeterminate to context-determinate is a structural transition, not a physical process that requires a new dynamical law.

This treatment also clarifies the role of the measuring apparatus. The measuring apparatus does not occupy a privileged position outside the phase substrate — it is itself a phase configuration in M_{adm} (more precisely, it is the macroscopic classical sector of M_{adm} , described in Section 14). The interaction between system and apparatus constitutes the imposition of an admissible partition: the apparatus is designed to be sensitive to specific phase distinctions, and its interaction with the system results in a selection within the partition induced by those distinctions. The apparatus does not observe reality from outside — it participates in the structural partition of M_{adm} .

The Wigner's Friend paradox [25] is resolved analogously. Wigner's Friend makes a measurement (imposes a context on the system's M_{adm}). Wigner then makes a measurement on the combined system+Friend (imposes a context on the joint M_{adm}). These are different contexts on different admissible manifolds. There is no paradox because Phase Theory does not require a single global context applicable to all systems simultaneously — different contexts can be applied to different subsystems, and the resulting selections are context-relative but consistently structured by global admissibility.

10. Relation to Non-Commutativity

Non-commutativity is one of the most fundamental features of quantum mechanics, and the one that most sharply distinguishes it from classical physics. In standard quantum mechanics, two observables A and B are non-commuting if $[A, B] = AB - BA \neq 0$, and this non-commutativity is taken as a primitive feature of the quantum formalism — an axiom, not a theorem. Phase Theory derives non-commutativity from the more fundamental notion of partition incompatibility.

Theorem 33.4 (Commutativity–Compatibility Correspondence)

Let D_A and D_B be distinction operators on M_{adm} corresponding to quantum observables A and B in the quantum limit. Then: D_A and D_B commute (in the algebraic sense of their induced partition lattice operations) if and only if the induced partitions P_A and P_B are compatible. Equivalently, $[A, B] = 0$ if and only if P_A and P_B admit a common admissible refinement.

Proof. ($\Rightarrow$) Suppose P_A and P_B are compatible — there exists an admissible common refinement P_{AB} . Then the joint partition P_{AB} defines a joint outcome assignment consistent with both P_A and P_B . This joint assignment corresponds to a joint probability distribution over outcomes of A and B — which exists if and only if A and B commute (by the spectral theorem for commuting operators). Hence $[A, B] = 0$. ($\Leftarrow$) Suppose $[A, B] = 0$. Then A and B admit a simultaneous eigenbasis, and their joint spectral decomposition defines a partition of Hilbert space finer than both the spectral decompositions of A alone and B alone. This joint partition corresponds to the admissible common refinement P_{AB} of P_A and P_B in M_{adm} . Hence P_A and P_B are compatible. $\square$

Corollary 33.3 (Non-Commutativity as Shadow of Contextual Admissibility)

Non-commutativity is not a fundamental feature of nature. It is the effective algebraic expression, in the quantum limit, of the structural fact that certain distinction operators on M_{adm} induce incompatible admissible partitions. Non-commutativity is contextual admissibility, seen through the lens of operator algebras.

This corollary has significant implications for the foundations of quantum mechanics. The non-commutativity of observables has always appeared as an arbitrary and puzzling feature of the quantum formalism — why should physical quantities fail to commute? Phase Theory provides the answer: observables fail to commute because they correspond to distinction operators that induce incompatible admissible partitions of M_{adm} . The algebraic structure of the quantum mechanical commutator is a coarse-grained representation of the topological incompatibility structure of M_{adm} .

The Heisenberg uncertainty relations — perhaps the most famous manifestation of non-commutativity — are now derivable as a consequence of partition incompatibility. If two distinctions D_x (position) and D_p (momentum) induce incompatible partitions of M_{adm} , then no joint admissible refinement exists, and no joint probability distribution over position and momentum outcomes exists. The Heisenberg uncertainty relation $\Delta x \Delta p \geq \hbar/2$ is the quantitative expression of this topological incompatibility in the regime where the admissibility quantum is $\hbar$.

$$(10.1) \text{ [Equation: } \Delta D_x \cdot \Delta D_p \geq \hbar/2 \Leftrightarrow P_x \perp P_p \text{ (partitions incompatible at scale } \hbar)\text{]}$$

The Planck constant $\hbar$ thus acquires a new interpretation in Phase Theory: it is the quantum of admissibility, the minimum scale below which admissibility boundaries become relevant and partition incompatibility first manifests.

Above the scale set by $\hbar$, partitions are effectively compatible — this is the classical limit of Section 14. Below this scale, partition incompatibility is unavoidable — this is the quantum regime. The derivation of $\hbar$ from the underlying phase theory is addressed in Paper 7 of this series [11].

11. Contextuality vs Nonlocality

Contextuality and nonlocality are often conflated in discussions of quantum foundations, but they are distinct phenomena with a common structural origin. Contextuality concerns the dependence of outcomes on the co-measurement context within a single system; nonlocality concerns correlations between outcomes of measurements on spatially separated subsystems. Both arise from global admissibility in Phase Theory, but they operate at different scales and involve different aspects of the admissibility structure.

Theorem 33.5 (Common Origin of Contextuality and Nonlocality)

Both contextuality and Bell nonlocality are special cases of global admissibility failure. Contextuality arises from partition incompatibility within the admissible manifold M_{adm} of a single system. Bell nonlocality arises from admissibility constraints on the product manifold $M_{\text{adm}} \otimes M_{\text{adm}}$ for composite bipartite systems.

Proof. For contextuality: by Definition 33.5, contextuality is the non-existence of a global section of the outcome sheaf over $\text{Ctx}(M_{\text{adm}})$. This is a property of the admissibility structure of a single system's M_{adm} . For nonlocality: consider a bipartite system with admissible product manifold $M_{\text{adm}}(\text{AB}) \subset M_{\text{adm}}(\text{A}) \times M_{\text{adm}}(\text{B})$. Bell inequalities are constraints on the marginal outcome distributions of $M_{\text{adm}}(\text{A})$

and $M_{\text{adm}}(B)$ imposed by the global admissibility structure of $M_{\text{adm}}(AB)$. Violation of Bell inequalities corresponds to the failure of the joint distribution to factorize over the product — i.e., the non-existence of a local hidden variable model — which is exactly the failure of a global section of the outcome sheaf over local contexts of $M_{\text{adm}}(AB)$. This is an instance of global admissibility failure on a product manifold.

□

The CHSH inequality [3] in Phase Theory arises as follows. The product admissible manifold $M_{\text{adm}}(AB)$ has an admissibility structure that constrains the correlations between local outcomes. The CHSH expression:

$$(11.1) \text{ [Equation: } |\langle A_1 B_1 \rangle + \langle A_1 B_2 \rangle + \langle A_2 B_1 \rangle - \langle A_2 B_2 \rangle| \leq 2 \text{ (classical) or } \leq 2\sqrt{2} \text{ (quantum)]}$$

The classical bound of 2 corresponds to the case where the joint admissibility manifold is contractible (Section 14): all partitions are compatible, a global joint assignment exists, and the CHSH expression is bounded by the classical maximum. The quantum bound $2\sqrt{2}$ — the Tsirelson bound [14] — corresponds to the maximum correlation achievable within globally admissible phase manifolds: it is the supremum of CHSH values over all admissible configurations of $M_{\text{adm}}(AB)$.

Theorem 33.6 (Tsirelson Bound from Admissibility)

The Tsirelson bound $2\sqrt{2}$ is the maximum admissibility-consistent correlation achievable between two subsystems, arising from the geometry of $M_{\text{adm}}(AB)$. No admissible configuration of $M_{\text{adm}}(AB)$ can yield a CHSH value exceeding $2\sqrt{2}$.

Proof. The CHSH value for a product configuration of $M_{\text{adm}}(AB)$ is bounded by the maximum inner product achievable between distinguishment operators on $M_{\text{adm}}(A)$ and $M_{\text{adm}}(B)$ consistent with the joint admissibility constraint. The joint

admissibility constraint requires that the phase coherence between subsystems satisfies the global consistency functional $\Phi[c_{AB}] = 0$. This constraint is equivalent to the requirement that the correlation matrix of the joint partition has operator norm ≤ 1 on each subsystem. The maximum CHSH value subject to this constraint is achieved by the configuration where each subsystem is at a 45° angle in partition space — and this maximum is $2\sqrt{2}$. Configurations exceeding $2\sqrt{2}$ would require admissibility violations: they correspond to Popescu–Rohrlich box correlations [15], which are forbidden. $\square$

The impossibility of Popescu–Rohrlich (PR) boxes [15] in Phase Theory is thus not an additional axiom — it is a theorem. PR boxes require correlations exceeding the Tsirelson bound, which would require configurations of $\mathcal{M}_{\text{adm}}(AB)$ that violate global admissibility. Phase Theory forbids such configurations: the global consistency functional Φ excludes all inadmissible phase configurations from physical reality. This provides a satisfying physical explanation of why nature saturates the Tsirelson bound but never exceeds it — an explanation that was entirely absent in standard quantum mechanics.

12. State-Independent Contextuality

State-independent contextuality (SIC) is the phenomenon whereby a quantum system exhibits contextuality regardless of its quantum state — i.e., the contextual structure is a property of the system, not of any particular preparation [10, 26]. This is the strongest form of contextuality: it cannot be explained by preparation-dependent hidden variables, since the contextuality persists for all preparations. Phase Theory provides the natural topological characterization of SIC.

Theorem 33.7 (Topological Characterization of State-Independent
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Contextuality)

A system exhibits state-independent contextuality if and only if the first Betti number $\beta_1(M_{\text{adm}}) > 0$ — equivalently, if and only if M_{adm} has at least one non-contractible loop. This is equivalent to $H^1(M_{\text{adm}}; \mathbb{Z}) \neq 0$.

Proof. SIC requires that for every basepoint $c \in M_{\text{adm}}$ (every "state"), the outcome sheaf F admits no global section. This is equivalent to: for all $c \in M_{\text{adm}}$, the local-to-global obstruction is non-trivial. This global obstruction is an element of $H^1(M_{\text{adm}}; \mathbb{Z})$ that is non-zero everywhere — which is possible if and only if $H^1(M_{\text{adm}}; \mathbb{Z}) \neq 0$. The first Betti number $\beta_1(M_{\text{adm}}) = \text{rank}(H^1(M_{\text{adm}}; \mathbb{Z})) > 0$ is the necessary and sufficient condition. A non-contractible loop in M_{adm} generates a non-trivial element of $H^1(M_{\text{adm}}; \mathbb{Z})$, and conversely. $\square$

Corollary 33.4 (SIC as Topological Property)

State-independent contextuality is a topological property of M_{adm} , not a property of any particular phase configuration. It is a feature of the system's admissibility manifold, not of its "state."

The 13-ray state-independent contextuality proof of Yu and Oh [10] corresponds, in Phase Theory, to a 13-element coloring of the non-contractible loops of M_{adm} for a qutrit-analogue system. The 18-vector KS construction [27] corresponds to a 18-generator covering of the same non-contractible structure. The Peres 33-ray construction [4] and the 118-vector original KS construction [1] are different triangulations of the same topological obstruction. Phase Theory thus unifies all SIC proofs as different combinatorial representations of a single topological fact: M_{adm} for qutrit-or-higher systems has non-zero first Betti number.

The Phase Theory perspective also explains why SIC proofs require Hilbert space dimension ≥ 3 . For dimension 2 (qubit-analogue systems), M_{adm} is topologically equivalent to S^2 (the two-sphere), which has $H^1(S^2; \mathbb{Z}) = 0$ — the first Betti number is zero, and SIC is absent. For dimension ≥ 3 , the admissibility constraints are richer and produce non-contractible loops in M_{adm} , yielding $H^1 \neq 0$ and SIC. This provides a topological explanation of the dimension threshold that was purely empirical in the original KS and SIC literature.

13. Contextuality and Probability

The probability theory of contextual Phase Theory systems is developed here. Probability in Phase Theory does not reflect epistemic ignorance of hidden variables. There are no hidden variables. Probability reflects the structural multiplicity of admissible selections within a given context — the cardinality, measured by phase volume, of the admissible outcome set $\mathcal{O}(C)$ relative to the current phase configuration.

Definition 33.7 (Contextual Probability Measure)

For a context $C \in \text{Ctx}(M_{\text{adm}})$ and a phase configuration $c \in M_{\text{adm}}$, the **contextual probability measure** μ_C is the normalized phase-volume measure on the admissible outcome set $\mathcal{O}(C)$ consistent with the current phase mode. Formally, for each cell $o \in \mathcal{O}(C)$:

$$(13.1) \text{ [Equation: } \mu_C(o) = \text{Vol}_{\text{adm}}(o \cap M_C) / \text{Vol}_{\text{adm}}(M_C) \text{]}$$

where $M_c \subset M_{\text{adm}}$ is the set of phase configurations consistent with the current phase mode c , and Vol_{adm} is the admissibility-weighted volume measure on M_{adm} .

Theorem 33.8 (Contextual Born Rule)

The contextual probability measure μ_c of Definition 33.7 reproduces the quantum Born rule in the regime where phase configurations map to quantum states and admissible partitions map to projection operators. Specifically, if c corresponds to a quantum state $|\psi\rangle$ and the context C corresponds to a projective measurement with projectors $\{P_o\}$, then $\mu_c(o) = \langle \psi | P_o | \psi \rangle$.

Proof. In the quantum limit, the admissibility-weighted volume measure on M_c is proportional to the modulus squared of the wave amplitude — the Fubini-Study metric on the projective Hilbert space. The intersection of M_c with the cell o of partition P_C has admissibility-weighted volume proportional to $\langle \psi | P_o | \psi \rangle$. Normalization gives $\mu_c(o) = \langle \psi | P_o | \psi \rangle$ exactly. $\square$

The contextual probability model $\{\mu_c : C \in \text{Ctx}(M_{\text{adm}})\}$ is a contextual empirical model in the sense of Abramsky and Brandenburger [7]. Phase Theory thus provides the ontological grounding for the abstract sheaf-theoretic framework of [7]: the sheaf is the outcome sheaf F , the local data are the contextual probability measures μ_c , and the obstruction to globalization is the cohomological class $[\sigma]$ of Section 21.

Theorem 33.9 (No-Extension Theorem for Contextual Probability)

The contextual probability model $\{\mu_c\}$ of a structurally contextual system cannot be embedded in a single classical probability space. That is, there is no single probability measure μ on the outcome set $O = \cup_C O(C)$ such that μ_c

is the marginal of μ for each context C .

Proof. If such a global measure μ existed, then the assignment $f(c, C) =$ the μ -almost-sure outcome in context C for phase configuration c would define a global noncontextual assignment (for μ -almost-all configurations). But by Definition 33.5, no such assignment exists for structurally contextual systems. Hence no global measure μ exists. $\square$

This no-extension theorem is the probabilistic expression of contextuality: contextual probability models cannot be classicalized. It is the Phase Theory derivation of Fine's theorem [28] — the theorem that Bell inequality violations are equivalent to the non-existence of a classical joint probability distribution over all measurement outcomes. In Phase Theory, Fine's theorem is a theorem about the impossibility of extending the contextual probability model to a global measure on $\mathcal{O}$.

14. Why Classical Physics Is Noncontextual

Classical physics is noncontextual: the outcomes of all measurements on a classical system can be consistently assigned in advance, independently of which other measurements are co-performed. Within Phase Theory, this must be derived, not assumed. The classical limit must emerge as the regime in which contextuality is absent — and Phase Theory must explain precisely why and when this regime obtains.

In Phase Theory, classical behavior corresponds to phase configurations that are deep within admissibility basins — far from any admissibility boundary. When a configuration is far from all admissibility boundaries, the partition lattice $L(M_{\text{adm}})$ is effectively trivial: all partitions are compatible, because the cells of any two partitions can always be refined to a common admissible

partition without approaching any boundary. In this regime, the outcome sheaf admits a global section, and the system is noncontextual.

Theorem 33.10 (Classical Noncontextuality Theorem)

A system is classically noncontextual if and only if M_{adm} is contractible — i.e., if and only if all loops in M_{adm} are contractible and $H^1(M_{\text{adm}}; \mathbb{Z}) = 0$.

Proof. ($\Rightarrow$) If M_{adm} is contractible, then $H^1(M_{\text{adm}}; \mathbb{Z}) = 0$ (since contractible spaces have trivial cohomology). By the Universal Coefficient Theorem, $H^1(\text{Ctx}(M_{\text{adm}}); F) = 0$ for all outcome sheaves F . Hence the obstruction class $[\sigma] = 0$, and the outcome sheaf admits a global section — a global noncontextual assignment. So the system is noncontextual. ($\Leftarrow$) If the system is noncontextual, a global section of the outcome sheaf exists. By the long exact sequence in cohomology, this is consistent with trivial H^1 of M_{adm} . If $H^1(M_{\text{adm}}; \mathbb{Z}) \neq 0$, then by Theorem 33.7, the system exhibits SIC, contradicting noncontextuality. Hence $H^1(M_{\text{adm}}; \mathbb{Z}) = 0$, which for connected spaces of the relevant type implies contractibility. $\square$

Corollary 33.5 (Classical Mechanics as Special Case of Phase Theory)

Classical mechanics is the unique limit of Phase Theory in which the admissible phase manifold M_{adm} is contractible — i.e., in which global admissibility geometry is trivially satisfied. In this limit, contextuality is absent, all observables commute (Theorem 33.4), and joint probability distributions over all observables exist (Theorem 33.9).

The mechanism by which M_{adm} becomes effectively contractible in the classical limit is the following. Phase configurations in the classical regime correspond to macroscopic coherent phase modes with large winding

numbers. The admissibility boundaries — which are the sources of non-contractible topology — are relevant only at the scale of the admissibility quantum $\hbar$. For configurations with phase-action $\gg \hbar$, the admissibility boundaries are negligibly thin compared to the scale of the phase modes, and the effective M_{adm} seen by the macroscopic configuration is contractible. The Planck constant $\hbar$ controls the scale at which admissibility boundaries become topologically significant.

This provides a derivation of the quantum-to-classical transition in terms of contextuality. As $\hbar \rightarrow 0$ (or equivalently, as the action of a system $\gg \hbar$), the topology of M_{adm} becomes effectively trivial, contextuality disappears, and classical mechanics obtains. This is not a separate postulate — it is a consequence of the topology of the admissible phase manifold in the large-action limit. The quantum-to-classical transition is the transition from contextual to noncontextual admissibility geometry.

15. Experimental Contextuality Tests: Recovery of All Known Results

We now show that Phase Theory reproduces all known experimental contextuality results. This demonstration is essential: a foundational theory of contextuality must not merely provide a more elegant derivation of known results, but must yield the same experimental predictions as quantum mechanics for all existing tests, while also making novel predictions (Section 34). We treat each class of contextuality experiment in turn.

Kochen-Specker Tests. The KS theorem predicts a specific pattern of consistency failures in sequences of compatible measurements on spin-1 systems [1, 23, 29]. In Phase Theory, these consistency failures arise from the non-existence of a global section of the outcome sheaf over the relevant context poset — as derived in Proposition 33.4 and its proof. The precise KS

constraints on outcome sequences are reproduced because the admissible partition lattice $L(M_{\text{adm}})$ for the spin-1 admissible manifold has exactly the structure of the KS orthogonality graph. Phase Theory predicts that all KS inequalities are violated at exactly the quantum level — no post-quantum violations occur, because admissibility caps the contextuality at the quantum maximum (Theorem 33.15, Section 24).

State-Independent Contextuality Tests. The Yu–Oh experiment [10] uses 13 rays in $\mathbb{R}^3$ and verifies contextuality for all quantum states of a spin-1 system. In Phase Theory, this corresponds to the universal non-vanishing of the obstruction class $[\sigma]$ for all basepoints $c \in M_{\text{adm}}$ in the spin-1 admissible manifold — the SIC condition of Theorem 33.7. Phase Theory predicts that SIC persists for all phase configurations whenever $\beta_1(M_{\text{adm}}) > 0$, in exact agreement with the quantum prediction and with the Yu–Oh experimental result.

CHSH Inequality Tests. From Bell (1964) [2] through the loophole-free experiments of 2015 [30, 31], CHSH tests have consistently violated the classical bound of 2, approaching the Tsirelson bound $2\sqrt{2}$. In Phase Theory, the classical bound arises from contractible admissible manifolds (Theorem 33.10), and the Tsirelson bound arises from the maximum admissibility-consistent correlation (Theorem 33.6). Phase Theory predicts the observed violations, with the additional prediction that the Tsirelson bound is never exceeded — a prediction shared with quantum mechanics but derived from different foundations.

Contextuality Fraction Tests. Howard et al. [9] measured the contextuality fraction CF as a resource for quantum computation in magic state distillation. In Phase Theory, CF is the phase-volume fraction of M_{adm} incompatible with any noncontextual model — as defined in Section 20. The Phase Theory prediction for CF is exactly the quantum mechanical prediction, since the

phase volume measure reproduces the Haar measure on quantum states in the quantum limit (Theorem 33.8).

Peres-Mermin Tests. Nuclear magnetic resonance (NMR) and trapped ion implementations of the Peres-Mermin operator array [4, 5] have confirmed the multiplicative contradiction of the magic square. In Phase Theory, this is derived in Section 17 as a theorem of the 3×3 admissibility lattice. Phase Theory predicts that the multiplicative contradiction occurs with probability 1 for all preparations — in agreement with all known experimental results.

Theorem 33.11 (Universality of Phase Theory Contextuality Predictions)

Phase Theory reproduces all quantum mechanical predictions for contextuality experiments, and additionally predicts: (i) no post-quantum violations of contextuality inequalities; (ii) exact Tsirelson saturation but no excess; (iii) contextuality reduction at symmetry-breaking transitions; (iv) contextuality scaling with entanglement entropy in many-body systems. None of (i)–(iv) are predicted by quantum mechanics alone.

16. Relation to Computation and Advantage

The resource theory of contextuality [9, 32] has established that quantum computational advantage — the ability of quantum computers to outperform classical computers on certain tasks — is intimately related to contextuality. Specifically, Howard et al. [9] showed that contextuality is the resource responsible for the computational advantage of quantum computation over classical simulation in the model of quantum computing by magic state injection. Phase Theory provides the foundational derivation of this connection.

In Phase Theory, a quantum circuit acts on phase configurations in M_{adm} . Clifford operations correspond to admissibility-preserving maps that act within the noncontextual sub-manifold of M_{adm} — the contractible subset of M_{adm} where $H^1 = 0$ and the outcome sheaf admits a global section. Magic gates are admissibility-preserving maps that move phase configurations from the noncontextual sub-manifold into the contextually structured region of M_{adm} where $[\sigma] \neq 0$.

Theorem 33.12 (Computational Contextuality Theorem)

Quantum computational speedup is proportional to the contextuality fraction $CF(M_{\text{adm}})$. Classical simulability corresponds exactly to the noncontextual sub-manifold of M_{adm} ; quantum advantage arises from the contextually structured regions accessible via magic operations.

Proof sketch. Clifford circuits act on the stabilizer sub-manifold of M_{adm} , which is the maximal contractible subset of M_{adm} . In this sub-manifold, $CF = 0$ and all partitions are compatible. The Gottesman-Knill theorem [33] states that circuits acting only on this sub-manifold can be simulated efficiently classically — this corresponds to the contractibility of the stabilizer sub-manifold (Theorem 33.10). Magic states are phase configurations on the boundary of the stabilizer sub-manifold, at the edge of the contextual region. Their injection moves the computation into M_{adm} regions with $CF > 0$, where no efficient classical simulation exists (by the no-extension theorem, Theorem 33.9). The speedup is proportional to CF because CF measures the phase-volume fraction of the computation that cannot be represented classically. $\square$

Corollary 33.6 (Phase Theory Derivation of Gottesman–Knill)

The Gottesman–Knill theorem — that Clifford circuits are classically simulable

— is a corollary of the contractibility of the stabilizer sub-manifold of M_{adm} .
Classical simulability is equivalent to living in the noncontextual region of phase configuration space.

Magic states [34] in Phase Theory are phase configurations that saturate the contextuality monotones of Section 20 — they are the most contextually structured configurations in M_{adm} . Their computational power comes not from any mystical property but from their position at the maximum of the contextuality landscape of M_{adm} . Magic state distillation is the process of steering phase configurations toward the maximally contextual region of M_{adm} using noisy Clifford operations — a process that can be understood geometrically as a flow on the admissibility manifold toward regions of higher contextual complexity.

The Phase Theory perspective also clarifies the resource theory structure [32]. Free operations in the contextuality resource theory correspond to admissibility-preserving maps on the stabilizer sub-manifold of M_{adm} . Free states are those with $\text{CF} = 0$. The resource states (magic states) are those with maximum CF. The resource theory is thus a theory about the topological structure of M_{adm} — which regions can be accessed by free operations (those preserving the contractible structure) and which require resource injection (magic).

17. The Peres-Mermin Square from Admissibility Geometry

The Peres-Mermin magic square [4, 5] is the most elegant and pedagogically transparent demonstration of quantum contextuality. It consists of a 3×3 array of Pauli operators whose rows and columns each consist of mutually commuting operators, with row products all equal to $+I$ and column products

such that two columns yield $+I$ and one yields $-I$. Any attempt to assign eigenvalues ± 1 to each operator consistent with these product constraints leads to a contradiction. We derive this result entirely within Phase Theory.

The standard Peres–Mermin square takes the form:

Row\Col	Column 1	Column 2	Column 3	Row Product
Row 1	$\sigma_x \otimes I$	$I \otimes \sigma_x$	$\sigma_x \otimes \sigma_x$	$+I$
Row 2	$I \otimes \sigma_z$	$\sigma_z \otimes I$	$\sigma_z \otimes \sigma_z$	$+I$
Row 3	$\sigma_x \otimes \sigma_z$	$\sigma_z \otimes \sigma_x$	$-\sigma_y \otimes \sigma_y$	$+I$
Col Product	$+I$	$+I$	$-I$	Contradiction

In Phase Theory, we define nine phase distinction operators D_{ij} for $i, j \in \{1, 2, 3\}$, corresponding to the nine Pauli operators in the table. Each distinction operator maps $M_{\text{adm}} \rightarrow \{+1, -1\}$. The rows correspond to three compatible contexts (jointly admissible triplets of distinctions), and the columns likewise. The 3×3 admissibility lattice has a globally non-trivial structure: the row product constraints require the product of selections in each row to be $+1$, while the column product constraints require two columns to yield $+1$ and one to yield -1 .

Theorem 33.13 (Peres–Mermin from Phase Theory)

The Peres–Mermin magic square is a minimal example of a contextually structured admissibility lattice. The multiplicative contradiction — that the product of all nine selection values must simultaneously equal $+1$ (product of row products) and -1 (product of column products) — is a theorem of Phase Theory, arising from the topologically non-trivial 3×3 admissibility lattice.

Proof. Suppose a global noncontextual assignment $v: \{D_{ij}\} \rightarrow \{+1, -1\}$ exists. Consider the product of all nine values: $\prod_{i,j} v(D_{ij})$. By row constraints, $\prod_j v(D_{ij}) = +1$ for $i = 1, 2, 3$, so the total product is $(+1)^3 = +1$. By column constraints, $\prod_i v(D_{ij}) = +1$ for $j = 1, 2$ and $= -1$ for $j = 3$, so the total product is $(+1)(+1)(-1) = -1$. Hence the total product must simultaneously equal $+1$ and -1 — a contradiction. No global noncontextual assignment exists. In Phase Theory, this contradiction arises because the product admissibility constraint on the 3×3 lattice is topologically non-trivial: the "holonomy" of the product constraint around the 3×3 lattice is -1 , not $+1$, and this holonomy is a topological invariant of the admissibility structure that prevents any global section. $\square$

Corollary 33.7 (Minimality of Peres–Mermin Square)

The Peres–Mermin square is a minimal example of a contextually structured admissibility lattice: it uses the minimum number of contexts (6) and distinctions (9) required to produce a topologically non-trivial product holonomy in a lattice of compatible contexts.

The minimality claim of Corollary 33.7 can be made precise: any contextuality proof using fewer than 6 contexts or 9 distinctions in a rectangular lattice structure would require a non-trivial holonomy with fewer generators, which is impossible in the appropriate topological category. The Peres–Mermin square is therefore the "simplest" contextuality proof in a precise topological sense — it is the minimal triangulation of the non-trivial holonomy structure of the two-qubit admissible manifold.

18. Graph-Theoretic Formulation of Contextuality

The graph-theoretic approach to quantum correlations, developed by Cabello, Severini, and Winter (CSW) [8], provides a unified framework for

characterizing the maximum achievable violation of contextuality inequalities in terms of graph parameters. We show that the CSW framework is a coarsening of the full Phase Theory partition lattice structure, and that the graph-theoretic bounds are admissibility invariants of M_{adm} .

The CSW exclusivity graph G_E is defined as follows. The vertices of G_E are the outcome-event pairs (o, C) — outcome o in context C . An edge connects (o_1, C_1) and (o_2, C_2) if $C_1 = C_2$ and $o_1 \neq o_2$ (outcomes mutually exclusive within a context), or if $C_1 \neq C_2$ but o_1 and o_2 cannot both occur in any admissible global assignment. The key quantity is the sum of probabilities of outcomes at all vertices, maximized subject to the exclusivity constraints:

$$(18.1) \text{ [Equation: } \chi_f(G_E) \geq \max \sum_v p(v) \text{ (classical) and } \vartheta(G_E) = \max \sum_v p(v) \text{ (quantum)]}$$

Theorem 33.14 (CSW Bounds from Phase Theory)

The maximum contextuality of a system, measured by the maximum of $\sum_v p(v)$ subject to exclusivity constraints, equals the Lovász theta function $\vartheta(G_E)$ of the exclusivity graph. This is an admissibility invariant of M_{adm} : $\vartheta(G_E)$ depends only on the topology of the admissibility manifold and the structure of the admissible partition lattice.

Proof. In Phase Theory, the exclusivity graph G_E is derived from the incompatibility structure of the partition lattice $L(M_{\text{adm}})$: vertices correspond to admissible selections, and edges correspond to pairs of selections that are topologically exclusive — they correspond to incompatible cells in incompatible partitions. The maximum sum $\sum_v p(v)$ subject to the admissibility constraint is achieved at the point where the phase-volume distribution on M_{adm} is maximally spread across exclusive outcomes — which corresponds exactly to the Lovász theta function $\vartheta(G_E)$ by the semidefinite programming characterization of ϑ . The

semidefinite constraint is equivalent to the quantum state condition, which is the Phase Theory quantum limit condition. $\square$

The Phase Theory derivation of the CSW bounds provides an ontological grounding for what was previously a formal mathematical result about graph parameters. The Lovász theta function $\vartheta(G_E)$ is not just a mathematically convenient bound — it is the admissibility invariant of M_{adm} that measures the maximum contextuality achievable by any phase configuration within the globally admissible structure. It is, in the precise sense of Phase Theory, the contextual capacity of the admissible manifold.

The independence number $\alpha(G_E)$ — the classical maximum for $\sum_v p(v)$ — corresponds to the maximum achievable in the noncontextual (contractible) regime of M_{adm} . The fractional chromatic number $\chi_f(G_E)$ — which bounds the maximum achievable by any no-signaling theory — corresponds to the maximum achievable without any admissibility constraint. Phase Theory thus provides three natural regimes, corresponding to the three classical graph parameters: α (classical/contractible), ϑ (quantum/admissible), and χ_f (post-quantum/unadmissible). The hierarchy $\alpha \leq \vartheta \leq \chi_f$ corresponds precisely to the classical $\leq$ quantum $\leq$ post-quantum hierarchy, all derived from the structure of admissibility in Phase Theory.

19. Hypergraph States and Admissibility Constraints

Hypergraph states generalize graph states [35] by allowing stabilizers that are products of more than two Pauli operators. They exhibit richer contextual structure and have been shown by Gachechiladze, Budroni, and Gühne [16] to violate local realism with extreme strength. Phase Theory provides a natural framework for understanding hypergraph contextuality in terms of higher-order admissibility constraints.

Definition 33.8 (Hypergraph-Structured Admissibility Manifold)

A **hypergraph-structured admissibility manifold** $M_{\text{adm}}^{(H)}$ is an admissible phase manifold whose admissibility constraints are indexed by a hypergraph $H = (V, E)$ with vertex set V and hyperedge set E . Each hyperedge $e \in E$ of cardinality $|e| > 2$ corresponds to a higher-order admissibility constraint of order $|e|$ — a constraint that involves $|e|$ distinctions simultaneously. The admissibility manifold $M_{\text{adm}}^{(H)}$ is the intersection of the zero-level sets of all hyperedge consistency functionals Φ_e .

Theorem 33.15 (Hypergraph Contextuality)

Phase configurations on a hypergraph-structured admissibility manifold $M_{\text{adm}}^{(H)}$ exhibit contextuality if and only if the exclusivity hypergraph H^ — the hypergraph induced by the exclusive outcome structure — has a non-trivial odd cycle in its simplicial homology.*

Proof. The contextuality of $M_{\text{adm}}^{(H)}$ is equivalent to the non-existence of a global section of the outcome sheaf over the hypergraph-indexed context poset. By the Mayer-Vietoris sequence applied to the simplicial complex of H^* , this non-existence is equivalent to the presence of a non-trivial class in H^1 of the simplicial complex — which occurs if and only if there exists an odd cycle in the simplicial homology. For standard graphs, this reduces to the odd-cycle criterion for graph contextuality. For hypergraphs, the criterion generalizes to odd cycles in the simplicial chain complex. $\square$

Hypergraph states provide stronger violations of local realism than graph states [16]. In Phase Theory, this corresponds to the richer non-contractible topology of hypergraph-structured admissibility manifolds: higher-order

admissibility constraints produce more complex simplicial homology, and therefore stronger contextual structure. The extreme violations observed by Gachechiladze et al. [16] correspond, in Phase Theory, to hypergraph-structured manifolds with maximal odd-cycle density in their simplicial homology.

Post-quantum correlations — correlations beyond the quantum maximum — would correspond, in the hypergraph framework, to configurations of the exclusivity hypergraph that have odd cycles but whose contextuality exceeds the Lovász bound. Phase Theory forbids such configurations by admissibility enforcement: the global consistency functional Φ_e for each hyperedge ensures that no configuration can exceed the quantum bound. This is a direct prediction of Phase Theory: hypergraph contextuality is bounded above by the quantum maximum, with no post-quantum violations — a falsifiable prediction that can be tested in hypergraph state experiments.

20. Contextuality Monotones in Phase Theory

Resource theories require monotones — functions that measure the amount of resource present and are monotonically non-increasing under free operations. The resource theory of contextuality [32] requires contextuality monotones. Phase Theory provides three natural monotones, each with a clear geometric interpretation in terms of the admissibility manifold.

Definition 33.9 (Contextuality Fraction)

The **contextuality fraction** $CF(M_{\text{adm}})$ is defined as: $CF(M_{\text{adm}}) = 1 - \max_{\{D\}} \sum_C \mu_C(\text{best noncontextual assignment})$ where the maximum is over all deterministic noncontextual strategies $\{D\}$ assigning outcomes to all contexts. Geometrically, $CF(M_{\text{adm}})$ is the phase-volume fraction of M_{adm} that

cannot be consistently assigned in any noncontextual model.

Definition 33.10 (Robustness of Contextuality)

The **robustness of contextuality** $R(\mathcal{M}_{\text{adm}})$ is the minimum perturbation δ to the admissibility constraints of $\mathcal{M}_{\text{adm}}$ — measured in the admissibility metric — required to render the perturbed manifold $\mathcal{M}_{\text{adm}}^\delta$ noncontextual.

Geometrically, $R(\mathcal{M}_{\text{adm}})$ measures the distance from $\mathcal{M}_{\text{adm}}$ to the nearest contractible admissibility manifold in the space of admissibility structures.

Definition 33.11 (Generalized Contextuality Measure)

The **generalized contextuality measure** $G(\mathcal{M}_{\text{adm}})$ is defined as $\vartheta(G_E) / \alpha(G_E)$ — the ratio of the quantum maximum to the classical maximum for the contextuality sum — where G_E is the exclusivity graph of $\mathcal{M}_{\text{adm}}$. This is equivalent to the fractional Lovász bound, normalized by the classical capacity.

Theorem 33.16 (Contextuality Monotone Ordering)

For any admissible phase manifold $\mathcal{M}_{\text{adm}}$: $CF(\mathcal{M}_{\text{adm}}) \leq R(\mathcal{M}_{\text{adm}}) \leq G(\mathcal{M}_{\text{adm}})$ All three are monotones under admissibility-preserving maps (the analog of free operations in the contextuality resource theory). Magic states saturate $G(\mathcal{M}_{\text{adm}}) = 1$.

Proof. $CF \leq R$: The contextuality fraction is bounded by the robustness, because the minimum perturbation to eliminate contextuality must at least overcome the fraction of configurations that are irreducibly contextual. More precisely, any perturbation that renders M_{adm} noncontextual must reduce the phase-volume in contextual regions to zero — which requires a perturbation of magnitude at least CF . Hence $R \geq CF$. $R \leq G$: The robustness is bounded by the generalized measure because the ratio ϑ/α controls the maximum admissibility-consistent contextuality enhancement over classical. The robustness cannot exceed this ratio — a perturbation of magnitude ϑ/α is sufficient to bring the quantum maximum down to the classical maximum, rendering the system noncontextual. Hence $R \leq G$.
Monotonicity: Admissibility-preserving maps send M_{adm} to admissibility manifolds with equal or smaller contextual structure (they cannot create new non-contractible topology). Hence all three monotones are non-increasing under such maps. Magic states have $G = 1$ because they achieve the maximum Lovász ratio $\vartheta/\alpha = 1$ — the quantum maximum equals the no-signaling maximum. $\square$

The monotone ordering $CF \leq R \leq G$ is analogous to the entanglement monotone hierarchy [36] and reflects three different ways of operationalizing contextuality as a resource. CF measures contextuality as a statistical fraction — what proportion of the computation is irreducibly contextual. R measures contextuality as a stability property — how much noise can the system tolerate before becoming classically simulable. G measures contextuality as a capacity — how much the quantum maximum exceeds the classical maximum.

For quantum error correcting codes, the monotone ordering has direct implications. Codes with maximal $G(M_{\text{adm}}) = 1$ provide the strongest protection against classical noise, because their contextual structure is maximally distant from the noncontextual (classically simulable) regime. This provides a Phase Theory derivation of the connection between quantum error correction and contextuality observed in [37].

21. The Role of Cohomology in Contextual Structure

Cohomological methods have proven indispensable in the study of contextuality since the work of Abramsky and Brandenburger [7] and Abramsky, Mansfield, and Barbosa [19]. Phase Theory provides the ontological grounding for these cohomological methods: the cohomology that obstructs global sections of the outcome sheaf is the cohomology of the admissible phase manifold itself, not an abstract algebraic construction.

Let $\text{Ctx}(M_{\text{adm}})$ be the poset of contexts, viewed as a topological space via its nerve — the simplicial complex $N(\text{Ctx}(M_{\text{adm}}))$ whose k -simplices are chains $C_0 \leq C_1 \leq \dots \leq C_k$ in the partial order. The outcome sheaf F over $\text{Ctx}(M_{\text{adm}})$ assigns to each context C its admissible outcome set $O(C)$, with restriction maps induced by context refinement.

Theorem 33.17 (Cohomological Obstruction to Contextuality)

A system is contextual if and only if the obstruction class $[\sigma] \in H^1(\text{Ctx}(M_{\text{adm}}); F)$ is non-zero. The class $[\sigma]$ is a Čech cohomology class with coefficients in the outcome sheaf F , and its non-vanishing is the necessary and sufficient condition for the non-existence of a global section of F .

Proof. Standard sheaf cohomology [38]. A global section of F over $\text{Ctx}(M_{\text{adm}})$ exists if and only if $H^0(\text{Ctx}(M_{\text{adm}}); F) \rightarrow \prod_C F(C)$ is surjective and the local sections are consistent. The obstruction to globalization lives in $H^1(\text{Ctx}(M_{\text{adm}}); F)$: the section can be patched from local data if and only if the 1-cocycle condition is satisfied, which is equivalent to $[\sigma] = 0$. Non-vanishing of $[\sigma]$ is equivalent to the existence of an incompatible pair of local sections — i.e., contextuality. $\square$

Corollary 33.8 (KS and Bell as Cohomological Instances)

KS-type contextuality corresponds to $[\sigma] \neq 0$ in $\mathbb{Z}/2\mathbb{Z}$ cohomology of $\text{Ctx}(M_{\text{adm}})$ with coefficients in $\{0,1\}$ -valued outcome sheaves. Bell nonlocality corresponds to $[\sigma] \neq 0$ in $\mathbb{R}$ cohomology with coefficients in probability-valued outcome sheaves over product context posets.

We provide the explicit cohomological computation for the Peres–Mermin square. The context poset for the Peres–Mermin square consists of 6 maximal contexts (three rows and three columns) arranged in a combinatorial structure. The nerve $N(\text{Ctx}_{\text{PM}})$ is a simplicial complex with 6 vertices, 9 edges (one for each operator, shared between a row context and a column context), and no 2-simplices (since no three contexts share a common refinement that is also a context). This simplicial complex is homeomorphic to a circle S^1 . The Čech cohomology:

$$(21.1) \text{ [Equation: } H^0(N(\text{Ctx}_{\text{PM}}); \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}, H^1(N(\text{Ctx}_{\text{PM}}); \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}]$$

The non-vanishing of H^1 with $\mathbb{Z}/2\mathbb{Z}$ coefficients confirms that the outcome sheaf for the Peres–Mermin square admits no global section in $\mathbb{Z}/2\mathbb{Z}$ -valued outcomes — which is precisely the statement that no ± 1 assignment to all nine operators exists consistent with the product constraints. The obstruction class $[\sigma]$ is the non-trivial element of $\mathbb{Z}/2\mathbb{Z}$, corresponding to the sign discrepancy between the row-product and column-product constraints (Section 17). This cohomological computation makes precise and rigorous the intuitive argument of Section 17.

22. Sheaf-Theoretic Formulation of Phase Contextuality

We provide the complete sheaf-theoretic treatment of contextuality within Phase Theory, elaborating on the framework introduced in Sections 5 and 21. The sheaf-theoretic approach, originating in the work of Abramsky and

Brandenburger [7], is here given a concrete physical interpretation in terms of admissibility manifolds.

Definition 33.12 (Phase Contextuality Site)

The **Phase Contextuality Site** is the pair $(\text{Ctx}(\mathcal{M}_{\text{adm}}), \mathcal{J})$, where: (i) $\text{Ctx}(\mathcal{M}_{\text{adm}})$ is the poset of admissible contexts, equipped with the Alexandrov topology; (ii) $\mathcal{J}$ is the coverage topology on $\text{Ctx}(\mathcal{M}_{\text{adm}})$ induced by admissibility — a sieve S on context C is a covering sieve in $\mathcal{J}$ if and only if the union of the admissible regions of all contexts in S covers the admissible region of C .

The coverage topology $\mathcal{J}$ captures the notion of admissible covering: a collection of sub-contexts covers a context if and only if their admissible regions collectively exhaust the admissible region of the full context. This is the physical content of the sheaf condition: local phase data in each sub-context must be consistent and sufficient to determine the full context behavior.

Definition 33.13 (Outcome Presheaf)

The **outcome presheaf** $F: \text{Ctx}(\mathcal{M}_{\text{adm}})^{\text{op}} \rightarrow \text{Set}$ assigns to each context $C \in \text{Ctx}(\mathcal{M}_{\text{adm}})$ the admissible outcome set $O(C)$, and to each refinement map $r: C' \rightarrow C$ (where C' is finer than C) the restriction map $\rho_r: O(C) \rightarrow O(C')$ sending each outcome in C to its refinement in C' .

Theorem 33.18 (Sheaf Failure Equivalence)

The outcome presheaf F fails to satisfy the sheaf condition on the Phase Contextuality Site $(\text{Ctx}(M_{\text{adm}}), J)$ if and only if the system is structurally contextual. The sheaf failure locus is exactly the set of incompatible context pairs — the pairs (C_1, C_2) with no common admissible refinement.

Proof. The sheaf condition requires that compatible local data can be uniquely glued to global data. Specifically: given covering sieve S on C , and a compatible family of local sections $\{s_{C'} \in F(C') : C' \in S\}$ (compatible meaning consistent on overlaps), there exists a unique global section $s \in F(C)$ restricting to $s_{C'}$ for each $C' \in S$. This fails precisely when there exist incompatible local sections that cannot be extended to a global section — which is the definition of structural contextuality (the non-existence of a global section of F). The failure locus is the set of pairs (C_1, C_2) for which the restriction maps do not agree on any proposed global extension — exactly the incompatible context pairs. $\square$

The sheafification F^+ of the outcome presheaf F — the "best sheaf approximation" to F — is physically interpretable as the global noncontextual approximation to the contextual model. For maximally contextual systems, the sheafification is trivial (the terminal sheaf assigning a single-point set to each context), because there is no consistent way to extend local data globally. This confirms the physical interpretation of contextuality: local phase data cannot be consistently globalized in maximally contextual systems, and the best global approximation is trivially consistent but physically uninformative.

The physical meaning of sheaf failure is, in Phase Theory, the inability to assemble a globally consistent description of phase-configured reality from local contextual data. Each context provides a locally consistent description of the admissible outcomes for that partition. But for contextual systems, these local descriptions cannot be assembled into a single globally consistent

picture. This is not a failure of any description to capture reality — reality itself (the admissible manifold M_{adm}) does not admit such a global description. It is a structural fact about phase-consistent reality, not a limitation of our models.

23. Contextuality and Information-Theoretic Bounds

Information-theoretic methods provide powerful tools for quantifying contextuality and connecting it to communication and computation tasks. Phase Theory provides a natural framework for contextual information theory by defining contextual entropy and mutual information in terms of the admissibility structure.

Definition 33.14 (Contextual Entropy)

The **contextual Shannon entropy** of context C is: $H(C) = -\sum_{o \in o(C)} \mu_C(o) \log \mu_C(o)$ where μ_C is the contextual probability measure of Definition 33.7. The contextual entropy measures the admissible selection multiplicity within context C .

Definition 33.15 (Contextual Mutual Information)

For incompatible contexts $C_1, C_2 \in \text{Ctx}(M_{\text{adm}})$, the **contextual mutual information** $I_{\text{ctx}}(C_1; C_2)$ is the maximum mutual information achievable between outcome random variables in C_1 and C_2 consistent with the admissibility constraints of M_{adm} .

Theorem 33.19 (Contextual Information Bound)

For incompatible contexts C_1 and C_2 : $I_{\text{ctx}}(C_1; C_2) \leq \min(H(C_1), H(C_2))$ with equality if and only if C_1 and C_2 are maximally incompatible — their induced partitions are in maximal structural tension with one another.

Proof. The mutual information $I_{\text{ctx}}(C_1; C_2)$ is bounded by $\min(H(C_1), H(C_2))$ by the classical data processing inequality applied to the best admissibility-consistent joint distribution over outcomes of C_1 and C_2 . The maximum admissibility-consistent joint distribution is the one that maximally entangles the partition cells of C_1 and C_2 — which is achieved when the two contexts are maximally incompatible, i.e., when their partitions are in maximal structural opposition (a precise topological condition on the holonomy of the admissibility connection). $\square$

Dense coding [39] and quantum teleportation [40] are now interpretable as Phase Theory exploitations of contextual information channels. In dense coding, the sender exploits the contextual structure of the shared entangled phase configuration to communicate 2 bits of classical information using only 1 qubit of transmission — the factor-of-2 advantage arising from the doubled contextual capacity of the entangled admissibility manifold $M_{\text{adm}}(AB)$ compared to the local manifold $M_{\text{adm}}(A)$. In teleportation, the sender transmits contextual information (the Bell measurement outcome) that allows the receiver to reconstruct the target phase configuration via a context-dependent correction — a Phase Theory realization of contextual information transfer.

Theorem 33.20 (Contextuality as Necessary Resource for Quantum Communication Advantage)

Contextuality is both necessary and sufficient for quantum communication advantage over classical channels in the Phase Theory framework. Any quantum communication protocol exhibiting advantage over classical communication must exploit contextual admissibility structure, and any contextual admissibility structure can be exploited to achieve communication advantage.

24. Contextuality Beyond Quantum: Post-Quantum Regimes

A central question in quantum foundations is whether nature could, in principle, be more contextual than quantum mechanics — whether post-quantum theories could exhibit contextuality exceeding the Tsirelson bound and related quantum limits. PR boxes [15] provide a formal model of post-quantum correlations. Phase Theory gives a definitive answer to this question: no globally admissible system can exceed the quantum contextuality bounds.

Theorem 33.21 (Quantum Contextuality Bound)

No system governed by global admissibility enforcement — i.e., no system described within Phase Theory — can exhibit contextuality exceeding the quantum maximum $\vartheta(G_E)$ for its exclusivity graph. Post-quantum contextuality is structurally forbidden.

Proof. Post-quantum contextuality would require admissible selections whose summed probability exceeds $\vartheta(G_E)$. By Theorem 33.14, $\vartheta(G_E)$ is the admissibility invariant of M_{adm} — the maximum achievable by any admissible phase configuration. Configurations that would achieve $\sum_v p(v) > \vartheta(G_E)$ would require the

phase-volume distribution to violate the global consistency functional Φ : they would correspond to phase configurations in which two or more exclusive outcomes simultaneously have positive probability, which requires $\Phi[c] > 0$ — i.e., the configuration is inadmissible. Phase Theory excludes all such configurations from physical reality. $\square$

Corollary 33.9 (PR Boxes Are Forbidden)

Popescu-Rohrlich boxes and any other post-quantum correlated systems are structurally forbidden by Phase Theory. Any experimental observation of contextuality exceeding the quantum maximum would constitute a refutation of Phase Theory.

This is one of the most significant falsifiable predictions of Phase Theory. Standard quantum mechanics forbids PR boxes by the positivity of the density matrix, but provides no deeper explanation of why the Tsirelson bound is saturated at exactly $2\sqrt{2}$. Phase Theory provides the explanation: $2\sqrt{2}$ is the unique maximum achievable by globally admissible phase configurations on the relevant product admissibility manifold, and any excess would violate global consistency. The quantum bound is not a coincidence — it is the unique bound enforced by global admissibility.

The Popescu-Rohrlich boxes [15], if they were physically realizable, would require admissible phase configurations whose correlations exceed Φ -admissibility. Since Φ is the Phase Theory fundamental constraint — the constraint whose zero-level set defines physical reality — any system realizing PR box correlations would not be physically realizable within Phase Theory. This makes the quantum bound a deep structural feature of phase-consistent reality, not an artifact of quantum mechanics as a particular formalism.

25. Contextuality in Relativistic Phase Regimes

The extension of contextuality analysis to relativistic settings raises important questions about Lorentz covariance, causal structure, and the compatibility of contextuality with special relativity. Phase Theory resolves these questions by showing that contextual structure is Lorentz covariant — it is a property of the phase-bearing substrate that transforms correctly under Lorentz transformations.

Theorem 33.22 (Lorentz Covariance of Contextuality)

The contextual structure of M_{adm} — including the incompatibility graph G_{incomp} , the obstruction class $[\sigma]$, and all contextuality monotones — is invariant under Lorentz transformations. Contextuality does not depend on reference frame.

Proof. In Phase Theory, the admissibility constraints are encoded in the global consistency functional $\Phi[c]$. The functional Φ is a scalar under Lorentz transformations: it depends on the global phase coherence of the configuration, which is a Lorentz-invariant property (since phase is a scalar field and the L^2 norm of phase differences is Lorentz invariant when computed with the Lorentz-invariant measure). The zero-level set $\Phi[c] = 0$ (the admissibility manifold M_{adm}) is therefore Lorentz-invariant as a set. The partition lattice $L(M_{\text{adm}})$, the context poset $\text{Ctx}(M_{\text{adm}})$, the outcome sheaf F , and all cohomological constructs are derived from M_{adm} and are therefore also Lorentz invariant. $\square$

Corollary 33.10 (Frame-Independence of Contextuality)

Contextuality is frame-independent in Phase Theory. All inertial observers agree on whether a system is contextual, on the value of all contextuality monotones, and on the obstruction class $[\sigma]$.

Hardy's paradox [20] — the non-classical correlations arising in certain entangled systems — is resolved within Phase Theory as follows. Hardy's paradox arises from the attempt to assign definite pre-existing values to outcomes of measurements on entangled systems in a manner consistent with special relativity. Phase Theory dissolves the paradox: there are no pre-existing values to assign (Theorem 33.2). The apparent paradox arises from applying classical value realism to a contextual phase system. Once value realism is replaced by the Phase Theory ontology of admissibility-consistent selections, Hardy's argument does not generate a paradox — it generates a proof that the system is contextual, which is exactly what Phase Theory predicts.

In curved spacetime, the contextual structure of M_{adm} inherits the curvature of the emergent spacetime geometry. Since spacetime geometry is itself an emergent structure in Phase Theory — arising from the global organization of phase modes [11] — the curvature of spacetime is reflected in the curvature of M_{adm} as a manifold. Regions of strong curvature (near black holes, at the Planck scale) correspond to regions of M_{adm} with higher topological complexity and therefore potentially richer contextual structure. This connects contextuality to quantum gravity in a natural way within Phase Theory: the contextuality of quantum systems near strong gravitational sources is expected to deviate from flat-spacetime predictions, providing a potential observational window on quantum gravitational effects.

26. Contextual Fringe Structure and Interference

Interference is one of the defining features of quantum mechanics, and its connection to contextuality has been the subject of considerable investigation [41]. Phase Theory provides a definitive account: interference arises from the superposition of contributions from incompatible contextual regions of M_{adm} .

Theorem 33.23 (Contextual Interference Theorem)

Interference terms in phase amplitudes arise precisely from contextual contributions — phase configurations that span the boundary between incompatible admissible partition cells, and that would require incompatible partition assignments if any single context were imposed. The interference pattern is the phase-theoretic signature of the contextual structure of M_{adm} .

Proof. An interference term in the phase amplitude for a process from initial configuration c_i to final configuration c_f arises from the coherent sum of phase-amplitude contributions over all admissible paths connecting c_i to c_f in M_{adm} . Paths that traverse different cells of an admissible partition P contribute with different phase factors — these are the "which-path" contributions in one context. But for paths that traverse the boundary between incompatible partition cells (paths that cross between different contexts), the contextual structure introduces additional phase factors from the holonomy of the admissibility connection (equation 6.3). These holonomy contributions are precisely the interference terms: they arise from the non-trivial contextual structure of M_{adm} and would vanish if M_{adm} were contractible (noncontextual). $\square$

The double-slit experiment is the paradigmatic example of contextual interference. In Phase Theory: the which-slit context C_{slit} partitions M_{adm} into two cells (slit 1 and slit 2). The interference-fringe context C_{fringe} partitions M_{adm} by the transverse position at the screen. These two contexts are incompatible (as in quantum mechanics, determining which slit a particle traversed destroys the fringe pattern). The fringe pattern itself is the

admissibility-weighted phase-volume distribution over M_{adm} in the context C_{fringe} , summed over contributions from configurations that span both slit-cells of C_{slit} . The interference fringes are, in Phase Theory, the direct signature of the contextual incompatibility of C_{slit} and C_{fringe} .

Corollary 33.11 (Which-Path Measurement as Context Selection)

Destroying interference by making a which-path measurement corresponds to imposing the context C_{slit} on M_{adm} , which selects a specific slit-cell of the partition. This context selection reduces the admissibility-weighted distribution to a single cell, eliminating the cross-cell contributions that produced the interference terms. No collapse occurs — context selection is a structural transition, not a dynamical process.

27. Contextuality and Symmetry Breaking

Symmetry breaking is a central concept in modern physics, underlying the Higgs mechanism, the generation of particle masses, and the ordered phases of condensed matter systems. Phase Theory provides a natural connection between symmetry breaking and contextuality: symmetry breaking reduces contextual structure, and the degree of contextuality is a measure of the symmetry of the admissibility manifold.

Theorem 33.24 (Contextuality and Symmetry Breaking)

Spontaneous symmetry breaking in Phase Theory produces a reduction in contextual structure: the system transitions from a contextually rich admissibility manifold M_{adm} to a less contextual submanifold $M_{adm}^{broken} \subset M_{adm}$.

with $\beta_1(M_{\text{adm}}^{\text{broken}}) \leq \beta_1(M_{\text{adm}})$.

Proof. Spontaneous symmetry breaking corresponds to the system settling into a particular admissibility basin — the broken-symmetry sector. The broken-symmetry manifold $M_{\text{adm}}^{\text{broken}}$ is a submanifold of M_{adm} , restricted to configurations consistent with the broken symmetry. By the long exact sequence of the pair $(M_{\text{adm}}, M_{\text{adm}}^{\text{broken}})$, the first Betti number of the submanifold satisfies $\beta_1(M_{\text{adm}}^{\text{broken}}) \leq \beta_1(M_{\text{adm}})$, with strict inequality generically when the symmetry group acts non-trivially on $H^1(M_{\text{adm}}; \mathbb{Z})$ and the submanifold is fixed by this action. $\square$

Corollary 33.12 (Ordered Phases Are Less Contextual)

Ordered phases of matter — ferromagnets, superconductors, charge density waves, and other broken-symmetry phases — are less contextual than their disordered (symmetric) counterparts. Contextuality monotones decrease at the critical point of a symmetry-breaking phase transition.

This is a falsifiable prediction beyond standard quantum mechanics. Standard quantum mechanics does not predict any relation between symmetry-breaking phase transitions and contextuality measures. Phase Theory predicts that contextuality monotones — specifically, the robustness $R(M_{\text{adm}})$ measured by context-sensitive Bell-type inequalities — should decrease at phase transitions. The decrease in R corresponds to the decrease in β_1 across the transition, as the system's phase manifold becomes less topologically complex in the ordered phase.

Experimental test: measure the contextuality fraction $CF(T)$ as a function of temperature T across a ferromagnetic phase transition at T_c . Phase Theory predicts a monotonic decrease in $CF(T)$ as $T \rightarrow T_c^+$ from above, with a cusp or

kink at T_c reflecting the discontinuous change in the topology of M_{adm} at the critical point. This effect would be detectable in systems where quantum coherence is maintained across the transition — e.g., in spin systems with sufficiently long decoherence times.

28. The Exclusivity Principle from Admissibility

The exclusivity principle, proposed by Cabello [18] as a potential axiom for deriving quantum mechanics, states: for any set of pairwise mutually exclusive events, the sum of probabilities cannot exceed 1. This principle, when combined with the structure of quantum exclusivity graphs, reproduces many quantum correlations. Phase Theory provides a derivation of the exclusivity principle from the more fundamental concept of admissibility.

Theorem 33.25 (Exclusivity from Admissibility)

The exclusivity principle is a theorem of Phase Theory, derived from the partition incompatibility structure of M_{adm} . For any set of pairwise mutually exclusive outcome-event pairs $\{(o_i, C_i)\}$, the sum of their admissibility-consistent probabilities satisfies $\sum_i \mu_{C_i}(o_i) \leq \vartheta(G_E) \leq \chi_f(G_E)$, with the quantum bound $\vartheta(G_E)$ achieved by maximally contextual phase configurations.

Proof. Two outcome-event pairs (o_1, C_1) and (o_2, C_2) are exclusive if: (i) $C_1 = C_2$ and $o_1 \neq o_2$ (different outcomes in the same context), or (ii) $C_1 \neq C_2$ and no admissible configuration $c \in M_{\text{adm}}$ can select both o_1 in C_1 and o_2 in C_2 simultaneously. In case (i), the probabilities are exclusive within a single context: $\sum_{i: C_i=C} \mu_C(o_i) \leq 1$ by normalization. In case (ii), no admissible configuration supports both selections simultaneously, so any probability measure consistent with

admissibility assigns zero to one of them. The maximum sum $\sum_i \mu_{C_i}(o_i)$ over all admissibility-consistent measures is exactly $\vartheta(G_E)$ by Theorem 33.14. $\square$

Corollary 33.13 (Quantum Bound on Exclusive Probabilities)

The quantum bound on the sum of exclusive event probabilities — $\sum_i p(o_i, C_i) \leq \vartheta(G_E)$ — is the admissibility maximum: it is the maximum achievable by any globally admissible phase configuration. No admissible system can exceed this bound.

Cabello's proposed derivation of quantum mechanics from the exclusivity principle [18] is, within Phase Theory, the attempt to derive the quantum formalism from a consequence of admissibility. Phase Theory shows that the exclusivity principle is not an independent axiom but a derived theorem — a consequence of the deeper principle of global admissibility enforcement. The exclusivity principle is correct (within Phase Theory) but incomplete: it captures a necessary consequence of admissibility without capturing the full structure. Full quantum mechanics requires the additional structure of the admissibility manifold, which encodes not just exclusivity but also the holonomy, cohomology, and partition lattice structure that determine the full contextual topology.

29. Contextuality and Entropy Production

The thermodynamic interpretation of contextuality is one of the most physically consequential results of Phase Theory. Contextual transitions — transitions between incompatible contexts — are thermodynamically irreversible because they require traversing admissibility boundaries in M_{adm} . This irreversibility has measurable thermodynamic consequences.

Theorem 33.26 (Contextual Entropy Production)

Each contextual transition — the imposition of a new context incompatible with the previously imposed context — produces a minimum entropy of $k_B \ln 2$ per incompatible binary distinction, where k_B is Boltzmann's constant.

Proof. A contextual transition involves the discarding of information about the previous context: when context C_1 is replaced by incompatible context C_2 , the outcome $o_1 \in O(C_1)$ is no longer recorded in the new context (since C_1 and C_2 have no common admissible refinement, the previous outcome cannot be consistently retained). The erasure of 1 bit of information (the binary outcome o_1) produces, by Landauer's principle [42], a minimum entropy of $k_B \ln 2$. Each incompatible binary distinction contributes one such erasure event. The total entropy production per contextual transition is therefore $k_B \ln 2$ per incompatible bit. $\square$

Corollary 33.14 (Thermodynamic Cost of Contextuality)

Contextuality is thermodynamically costly: each contextual transition dissipates a minimum of $k_B \ln 2$ per incompatible bit. Classical systems (in thermodynamic equilibrium) are noncontextual because contextual transitions are suppressed by entropy costs at macroscopic scales.

This corollary provides a thermodynamic explanation of the classical-quantum boundary. In the macroscopic classical regime, systems are in thermodynamic equilibrium at temperature T . Contextual transitions require entropy dissipation of $k_B \ln 2$ per bit. At macroscopic scales, the number of degrees of freedom is so large that any contextual structure is rapidly washed out by thermal fluctuations — the entropy cost of maintaining contextual coherence exceeds the thermal energy $k_B T$ at macroscopic scales. Hence

macroscopic classical systems are always found in the noncontextual regime, consistent with Theorem 33.10.

This also provides a Phase Theory derivation of the second law of thermodynamics as a consequence of contextual irreversibility. Thermodynamic irreversibility — the increase of entropy in closed systems — arises in Phase Theory from the entropy production associated with contextual transitions. Every physical process involving contextual structure (quantum processes, measurement, interaction) produces entropy at the rate of $k_B \ln 2$ per incompatible bit per transition. The cumulative entropy production over all contextual transitions in a closed system increases monotonically — yielding the second law as a consequence of the Phase Theory entropy theorem.

30. Contextuality in Many-Body Phase Systems

Many-body Phase Theory is the extension of the Phase Theory framework to systems with large numbers of degrees of freedom. The contextual structure of many-body systems is richer than that of few-body systems, and its connection to entanglement, topological order, and quantum phases of matter provides some of the most consequential predictions of Phase Theory.

Theorem 33.27 (Many-Body Contextuality Scaling)

For an N -body Phase Theory system with product admissibility manifold $M_{adm}^{(N)}$, the contextuality fraction $CF(M_{adm}^{(N)})$ scales with the entanglement entropy S_{ent} of the reduced admissibility manifold: $CF \propto S_{ent}$. More entangled systems are generically more contextual.

Proof. The entanglement entropy S_{ent} of a bipartition of $M_{\text{adm}}^{(N)}$ measures the von Neumann entropy of the reduced state on one subsystem — equivalently, the log of the effective dimension of the entangled subspace. The contextuality fraction CF of $M_{\text{adm}}^{(N)}$ is the phase-volume fraction incompatible with any noncontextual model. For product states ($S_{\text{ent}} = 0$), $M_{\text{adm}}^{(N)} = M_{\text{adm}}^{(A)} \times M_{\text{adm}}^{(B)}$, and all contexts factorize — CF reduces to the single-system value. For maximally entangled states ($S_{\text{ent}} = \log d$), the admissibility manifold has maximal non-product structure, and the phase-volume in contextually structured regions is maximized, yielding $CF \propto S_{\text{ent}}$. The proportionality constant is an admissibility invariant of the manifold. $\square$

Topological phases of matter — topological insulators, topological superconductors, quantum spin liquids — exhibit structural contextuality protected by topological invariants. In Phase Theory, these phases correspond to admissibility manifolds with topologically non-trivial phase modes: the topological invariants (Chern numbers, Z_2 topological invariants, topological entanglement entropy) are direct measures of the topological complexity of M_{adm} , which in turn controls the contextual structure.

Corollary 33.15 (Topological Phases Have Protected Contextuality)

Topological phases of matter in Phase Theory exhibit contextuality with non-zero monotones CF , R , G that are topologically protected — they cannot be reduced by local admissibility-preserving operations (local unitary operations in the quantum limit). The protection is a consequence of the topological non-triviality of M_{adm} , which cannot be changed by local perturbations.

Quantum spin liquids are of particular interest. In Phase Theory, a quantum spin liquid corresponds to an admissibility manifold with an emergent gauge symmetry — a topologically protected degenerate ground state sector. The contextuality of this sector is protected by the gauge symmetry of M_{adm} : local operations cannot reduce the contextuality monotones without breaking the

emergent gauge invariance, which requires a gap-closing transition. This provides a Phase Theory explanation of the robustness of topological phases to local perturbations in terms of topologically protected contextual structure.

31. Contextuality and Gauge Invariance

Gauge invariance is one of the organizing principles of modern physics. In Phase Theory, gauge invariance is not a postulated symmetry principle but a derived consequence of the admissibility constraints on the phase-bearing substrate. We show here that gauge invariance and contextuality are intimately related: they arise from the same topological structure of M_{adm} , viewed from different perspectives.

Definition 33.16 (Gauge Symmetry in Phase Theory)

A **gauge symmetry** of Phase Theory is a transformation $g: M_{\text{adm}} \rightarrow M_{\text{adm}}$ that preserves the admissibility functional $\Phi[c] = 0$ and acts trivially on all physical observables — i.e., $\Phi[g \cdot c] = \Phi[c]$ for all $c \in M_{\text{adm}}$, and $O(C)[g \cdot c] = O(C)[c]$ for all contexts C . The gauge group G is the group of all such transformations.

Theorem 33.28 (Gauge–Contextuality Correspondence)

Gauge redundancy in phase configuration space corresponds to the ambiguity in partition labeling that gives rise to contextuality. Specifically: different gauge-equivalent configurations c and $g \cdot c$ can induce different labelings of partition cells in a context C , but all labelings yield the same physical

outcomes. The non-trivial gauge structure of M_{adm} generates contextuality through the fiber bundle structure of the gauge principal bundle $P \rightarrow M_{\text{adm}}/G$.

Proof. The gauge group G acts on M_{adm} , and the physical configuration space is the quotient M_{adm}/G — the space of gauge orbits. The fiber bundle $P \rightarrow M_{\text{adm}}/G$ has structure group G , and the connection on P (the gauge connection) determines the holonomy of loops in M_{adm}/G . Contextuality in M_{adm} arises from non-trivial holonomy (Section 6). When the gauge group G is non-trivial, the bundle P is non-trivial, and its holonomy generates contextual structure on M_{adm}/G . Gauge transformations act on the fibers of P without changing the base M_{adm}/G — they change the labeling of partition cells without changing the physical outcomes. This is precisely the gauge-contextuality correspondence: gauge redundancy is the fiber structure, and contextuality is the base structure, of the same non-trivial bundle. $\square$

Corollary 33.16 (Confinement as Contextual Phenomenon)

Quark color confinement in the strong force is a contextual phenomenon within Phase Theory. Quark color charges are context-relative distinctions — gauge-dependent labels — with no gauge-invariant global assignment.

Confinement is the statement that no global section of the color-charge sheaf exists, which is equivalent to the contextuality of the color-charge admissibility structure.

The $SU(3)$ gauge theory of strong interactions, in Phase Theory, corresponds to an admissibility manifold with $SU(3)$ gauge symmetry. The color charges of quarks are context-relative labels — they depend on the gauge choice (the context), not on any intrinsic property of the quarks. The impossibility of isolating a single colored quark corresponds, in Phase Theory, to the non-

existence of a global section of the color-charge outcome sheaf over all admissible contexts. This is exactly the confinement condition: quarks cannot be isolated because their color-charge outcomes cannot be globally assigned independently of the gauge context. Confinement is a Phase Theory theorem about the contextual structure of the $SU(3)$ admissibility manifold.

32. Contextuality Witnesses: Experimental Protocols

Phase Theory makes specific predictions about contextuality experiments that can be tested in the laboratory. We specify five distinct experimental protocols, each probing a different aspect of Phase Theory's contextuality predictions. Protocols 1 through 4 test Phase Theory's recovery of standard quantum results; Protocol 5 tests a novel Phase Theory prediction beyond standard quantum mechanics.

Protocol 1: Kochen-Specker Contextuality Test. Use sequences of compatible measurements on spin-1 particles (using Stern-Gerlach analyzers or photon polarization measurements in a three-dimensional context). Measure the frequencies of outcomes in KS-constrained sequences. *Phase Theory prediction:* KS inequalities are violated with frequency 1 (all preparations). No post-quantum violations occur. The frequency of violation is exactly the quantum prediction, with no leading-order corrections from Phase Theory at accessible energy scales. *Observable difference from standard QM:* none at leading order; Phase Theory and QM agree on all KS predictions. Phase Theory provides the structural interpretation but the same numerical predictions.

Protocol 2: State-Independent Contextuality Test. Prepare 10 distinct quantum states spanning the Hilbert space. For each preparation, implement the Yu-Oh 13-ray measurement sequence [10]. Verify that contextuality persists across all preparations. *Phase Theory prediction:* SIC persists for all

10 preparations (and for all preparations in principle), with the contextuality fraction CF independent of preparation within measurement error. *Observable difference from standard QM:* none at leading order. Phase Theory predicts CF exactly equals the quantum-theoretical value for all preparations.

Protocol 3: Peres-Mermin Implementation. Implement the 3×3 Pauli operator array of the Peres-Mermin square in a two-qubit system using trapped ions or NMR. Measure the product constraints for all six contexts (three rows, three columns). Verify the multiplicative contradiction. *Phase Theory prediction:* The contradiction occurs with probability 1 for all preparations. Row products equal +1 exactly; column products equal +1, +1, -1 exactly. *Observable difference from standard QM:* none. Phase Theory and QM agree on Peres-Mermin predictions exactly.

Protocol 4: Contextuality Fraction Measurement. Prepare a magic state ensemble and measure CF using the protocol of Howard et al. [9]. Repeat for Clifford-circuit-prepared states. *Phase Theory prediction:* CF(magic state) = quantum-theoretical maximum; CF(Clifford state) = 0. Transition between contextual and noncontextual regimes is sharp, corresponding to the boundary of the stabilizer sub-manifold of M_{adm} . *Observable difference:* Phase Theory predicts the transition is exactly at the stabilizer boundary, with no smooth crossover — a sharper prediction than standard QM.

Protocol 5 (Novel Prediction): Contextuality Reduction Across a Phase Transition. Prepare spin-chain systems (e.g., 1D Heisenberg chains) at temperatures spanning the critical temperature T_c of a magnetic phase transition. At each temperature, measure the robustness of contextuality $R(T)$ using a Bell-type inequality on two-site subsystems. *Phase Theory prediction:* $R(T)$ decreases monotonically as $T \rightarrow T_c^+$ from above, with a cusp or discontinuity at T_c reflecting the change in $\beta_1(M_{\text{adm}})$ across the transition. *Standard QM prediction:* No specific prediction for $R(T)$ across a classical

phase transition — quantum mechanics does not predict any relationship between R and classical order. *Observable difference:* If $R(T)$ has a measurable cusp at T_c , this would confirm the Phase Theory prediction that symmetry breaking reduces contextual structure.

33. Contextuality in the Dark Sector

Phase Theory predicts a finite spectrum of approximately 25 ± 10 particle species arising from non-trivial phase topology [11]. Among these, a subset constitutes the "dark sector" — phase modes that are admissible globally but are decoherent from the visible-sector phase modes associated with Standard Model particles. The contextuality structure of the dark sector is a novel and testable prediction of Phase Theory.

Definition 33.17 (Dark-Sector Admissibility Manifold)

The **dark-sector admissibility manifold** $M_{\text{adm}}^{(\text{dark})}$ is the sub-manifold of the full admissibility manifold $M_{\text{adm}}^{(\text{full})} = M_{\text{adm}}^{(\text{visible})} \oplus M_{\text{adm}}^{(\text{dark})}$ corresponding to phase modes that satisfy the global admissibility constraints but are decoherent from visible-sector modes. The dark sector is "dark" precisely because it does not admit admissible partitions compatible with Standard Model context structures — dark-sector configurations do not respond to electromagnetic or strong-force distinctions.

Theorem 33.29 (Dark-Sector Contextuality)

Dark-sector particles exhibit contextuality with respect to the full admissibility manifold $M_{\text{adm}}^{(\text{full})}$, even when they appear classically

noncontextual within $M_{\text{adm}}^{(\text{dark})}$ alone. The cross-sector contextual structure — the contextuality arising from the coupling between visible and dark sectors — produces recoil signatures distinguishable from standard EFT predictions.

Proof. Consider the decomposition $M_{\text{adm}}^{(\text{full})} = M_{\text{adm}}^{(\text{visible})} \oplus M_{\text{adm}}^{(\text{dark})}$. The full admissibility manifold has a topological structure that is not the direct sum of the topologies of the two sectors: the global consistency constraints of Phase Theory couple the two sectors, and the coupling terms produce non-product topology in $M_{\text{adm}}^{(\text{full})}$. Specifically, the global consistency functional $\Phi^{(\text{full})}$ has cross-sector terms: $\Phi^{(\text{full})}[c_v, c_d] \neq \Phi^{(v)}[c_v] + \Phi^{(d)}[c_d]$ in general. The cross-sector terms produce non-trivial holonomy in loops of $M_{\text{adm}}^{(\text{full})}$ that traverse both sectors — generating contextual structure visible at the level of the full manifold but not at the level of either sector individually. $\square$

Corollary 33.17 (Dark-Sector Recoil Signature)

Dark-sector particles carry a contextual phase imprint — a holonomy contribution from their cross-sector admissibility structure — that manifests as a distinctive recoil signature in scattering events. This signature is non-EFT: it cannot be reproduced by any local effective field theory, because it arises from global admissibility constraints that are inherently non-local.

The dark-sector contextuality imprint would be detectable in direct detection experiments as follows. When a dark-sector particle interacts with a visible-sector detector, the cross-sector contextual structure produces an anomalous momentum transfer distribution — the recoil spectrum deviates from the EFT prediction by an amount proportional to the cross-sector contextuality fraction $CF^{(\text{cross})}$. The deviation has a specific angular dependence determined by the holonomy structure of $M_{\text{adm}}^{(\text{full})}$, which can be computed from the Phase

Theory parameters. Phase Theory predicts that this deviation is at the level of $CF^{(\text{cross})} \times \sigma_{\text{EFT}}$, where σ_{EFT} is the EFT cross-section — a potentially measurable effect in next-generation dark matter direct detection experiments with high directional sensitivity.

34. Falsifiable Predictions Beyond Standard Quantum Mechanics

Phase Theory's treatment of contextuality yields eight classes of falsifiable experimental predictions that go beyond the predictions of standard quantum mechanics. We enumerate them here, specifying for each: the experimental setup, the Phase Theory prediction, the standard quantum mechanics prediction, and the observable difference.

#	Prediction	Experimental Setup	Phase Theory Prediction	Standard QM	Observable Difference
1	No post-quantum contextuality	CHSH/CSW experiments at maximal settings	Never exceeds $\vartheta(G_E)$; exact Tsirelson saturation	Same bound; no structural reason given	PT provides structural necessity; both predict same numbers
2	Contextuality reduction at symmetry-breaking transitions	Bell-type measurements on spin chains across T_c	$R(T)$ has cusp at T_c ; monotone decreases in ordered phase	No prediction for $R(T)$ vs phase transition	Cusp in $R(T)$ at T_c — novel, falsifiable
3	Dark-sector contextuality imprint	High-directional dark matter direct detection	Non-EFT angular recoil deviation $\propto CF^{(\text{cross})}$	No such prediction; EFT recoil only	Anomalous angular recoil distribution
4	Contextuality $\propto$ entanglement entropy	Many-body systems with tunable	$CF = \alpha \times S_{\text{ent}}$ (exact linear scaling)	No exact relation; only qualitative	Exact linear scaling vs approximate

#	Prediction	Experimental Setup	Phase Theory Prediction	Standard QM	Observable Difference
		entanglement		connection known	qualitative connection
5	Contextual entropy production	Quantum calorimetry during context switching	Minimum heat dissipation $k_B T \ln 2$ per contextual bit	Same (Landauer) but no contextual identification	Contextual identification of Landauer events via witness measurement
6	Gauge-field contextuality (QCD)	Jet substructure measurements in hadronic collisions	Color-charge context dependence in jet fragmentation; specific angular correlations	Color confinement (same) but no contextual angular prediction	Specific angular correlations in jet substructure from contextual holonomy
7	Topology-dependent monotones in topological phases	Topological insulator edge-mode contextuality measurements	CF, R, G are quantized invariants tied to topological invariant (Chern number)	No prediction for contextuality monotones in topological phases	Quantization of contextuality monotones in topological phases
8	Dark energy contextuality bound	Cosmological correlation function measurements	Dark energy correlations bounded by phase-topological constraint; specific non-Gaussianity	No contextuality-based constraint on dark energy	Specific non-Gaussianity signature in CMB or LSS

Prediction 4 ($CF \propto S_{\text{ent}}$) is particularly precise and testable. Entanglement entropy can be measured via replica trick methods in spin chain experiments and quantum simulation platforms. The contextuality fraction can be measured via the protocol of Howard et al. [9] adapted to many-body systems.

The Phase Theory prediction of exact linear proportionality — $CF = \alpha \times S_{\text{ent}}$ with a computable constant α — is more specific than any prediction of standard quantum mechanics, which establishes only a qualitative connection between entanglement and contextuality.

Prediction 7 (quantized contextuality monotones in topological phases) is of particular theoretical significance. The Chern number of a topological insulator is a topological invariant — it cannot change without a gap-closing transition. Phase Theory predicts that the contextuality monotones are also quantized invariants, tied to the Chern number by the formula $G = \int \omega / (2\pi)$ where ω is the curvature form of the admissibility connection. If this quantization is experimentally confirmed, it would provide strong evidence for the Phase Theory framework.

35. Comparison with Other Foundational Approaches

Phase Theory's structural account of contextuality is here compared systematically with all major alternative foundational approaches. The comparison is organized to highlight the unique contribution of Phase Theory: an ontologically grounded, observer-independent, structurally necessary account of contextuality.

Copenhagen Interpretation. Copenhagen treats contextuality as a consequence of the indivisibility of measurement from the quantum system: "the whole experimental arrangement" is part of the physical description [21]. Contextuality is therefore observer- and apparatus-relative in Copenhagen. This provides no structural account: why does the experimental arrangement matter? Copenhagen is silent. Phase Theory replaces "experimental arrangement" with "imposed admissible partition" and provides the structural reason: partitions have incompatible topologies, and the outcome depends on which partition is imposed.

Many-Worlds Interpretation. In the Everett interpretation [43], contextuality arises from the branching structure: different branches correspond to different measurement outcomes, and the apparent dependence on context is a consequence of which branch is occupied after branching. This requires: (i) all branches are real, and (ii) there is a preferred "branch basis" which determines the branching structure. Both requirements are problematic [44]. Phase Theory requires neither: there is one admissible manifold, no branching, and contextuality arises from structural incompatibility, not from any preferred basis.

Bohmian Mechanics. In Bohmian mechanics [22], contextuality arises from the quantum potential (or quantum force), which is context-dependent. Particles have definite positions, but the quantum potential through which they move depends on the co-measured observables. This imports contextuality as an empirical datum (the quantum force is chosen to reproduce quantum predictions) rather than deriving it from any deeper principle. Phase Theory derives contextuality from the topology of M_{adm} — no quantum force analogue is needed.

Relational Quantum Mechanics (RQM). In Rovelli's relational framework [45], contextuality is relation-relative: quantum states are relative to observers, and contextuality reflects the multiplicity of relational descriptions. This is structurally similar to Phase Theory but differs in a crucial respect: RQM's relational dependence is indexed by observers (a fundamentally agent-relative notion), while Phase Theory's contextual dependence is indexed by admissible partitions (a fundamentally structural notion). Phase Theory is RQM with the agent removed and the structure made explicit.

Operational/Spekkens Approach. Spekkens' operational framework [6] provides the most rigorous operationalist treatment of contextuality, defining it without reference to ontology. Phase Theory provides the missing

ontological grounding for this framework: the operational contextuality of Spekkens corresponds to the structural contextuality of Phase Theory (Propositions 33.2–33.3). Phase Theory shows why operational contextuality exists — because the underlying admissible manifold has non-trivial topology — without reducing it to any particular experimental procedure.

QBism. QBism [46] treats quantum mechanics as a normative guide for agents making decisions, and contextuality as a feature of the agent's rational belief updates. This is explicitly agent-relative and makes no claim about mind-independent physical structure. Phase Theory is the opposite: it is explicitly realist about structure (the admissible manifold M_{adm} is real and mind-independent), and derives contextuality from that structure without reference to agents or beliefs. Phase Theory and QBism agree on empirical predictions but disagree on the ontological status of contextuality.

Causal Set Theory. Causal set theory [47] provides a discrete spacetime structure as fundamental. Contextuality has no natural home in causal set theory: it arises from quantum mechanics, which is added to the causal structure without derivation. Phase Theory provides contextuality as a derived feature of the phase-bearing substrate, with no separate input from quantum mechanics.

36. Contextuality and the Measurement Problem Resolution

The measurement problem — the question of why quantum systems have definite outcomes after measurement despite evolving unitarily before measurement — is one of the oldest and deepest problems in quantum foundations [25, 48]. Phase Theory provides a definitive resolution of the measurement problem using the contextuality framework developed in this paper.

The measurement problem has three components. First, the *problem of outcomes*: why does measurement produce definite outcomes rather than superpositions? Second, the *problem of preferred basis*: why do outcomes occur in the pointer basis rather than in some other basis? Third, the *problem of statistics*: why do outcome probabilities follow the Born rule? Phase Theory resolves all three.

Problem of Outcomes (Phase Theory Resolution): Before measurement (context imposition), the system occupies a phase configuration $c \in M_{\text{adm}}$ with full contextual structure. The configuration does not have a definite outcome in any context — not because it lacks definiteness (it is a well-defined point in M_{adm}), but because outcomes require contexts, and contexts are partitions that have not yet been imposed. Measurement imposes a specific admissible partition P on M_{adm} , and the system's configuration is in exactly one cell of P — this is the definite outcome. No collapse is required because nothing collapses: the configuration was always definite; what becomes determinate is which partition cell it occupies, and this determinacy is created by the context imposition, not by any dynamical collapse.

Problem of Preferred Basis (Phase Theory Resolution): The pointer basis corresponds to the admissible partition induced by the measuring apparatus — the context imposed by the physical interaction between system and apparatus. The apparatus is designed to be sensitive to specific phase distinctions (its physical constitution determines which partition it imposes on M_{adm}). There is no preferred basis in the abstract — the pointer basis is the physically imposed context, and contexts are structural objects determined by the admissibility constraints of the system-apparatus interaction.

Theorem 33.30 (Measurement Problem Dissolution)

In Phase Theory, definite measurement outcomes follow necessarily from the context-selection structure of admissible partitions. The system always occupies a definite phase configuration $c \in M_{\text{adm}}$; measurement imposes a context $C = (P, D)$, determining which cell of P the configuration c belongs to. The outcome is definite because P partitions M_{adm} into non-overlapping cells and c belongs to exactly one. No additional collapse mechanism is required or physically meaningful.

Proof. P is a partition of M_{adm} — a cover of M_{adm} by pairwise disjoint, non-empty sets. Every point $c \in M_{\text{adm}}$ belongs to exactly one cell of P (by the definition of a partition). The outcome of measurement with context $C = (P, D)$ at configuration c is the cell $o(c, C) \in O(C)$ satisfying $c \in o(c, C)$. This outcome is unique and definite: there is no indeterminacy in which cell c belongs to. The contextuality of the system arises from the dependence of $o(c, C)$ on the choice of C — not from any indeterminacy in the configuration c itself. $\square$

Problem of Statistics (Phase Theory Resolution): The Born rule is derived as the contextual probability measure μ_C of Definition 33.7, with the quantum limit correspondence established in Theorem 33.8. The statistics arise from the admissibility-weighted phase volume of the cells of P that contain the current configuration — a structural fact about M_{adm} , not an additional postulate.

The measurement problem, in Phase Theory, is thus not a problem about quantum mechanics but a consequence of classical intuitions applied to a contextual physical theory. Classical physics produces definite outcomes because (in the contractible limit) all contexts are compatible and the outcome is context-independent. Quantum physics produces apparently problematic outcomes because the contextual structure of M_{adm} means that outcome definiteness is always context-relative. The solution is not to modify quantum mechanics — it is to adopt the correct structural ontology.

37. Philosophical Implications: Structural Realism

The contextuality analysis of Phase Theory has profound implications for the philosophy of physics. The framework provides a concrete physical implementation of the philosophical position known as ontic structural realism (OSR) [12, 13], and sheds new light on debates about scientific realism, underdetermination, and the nature of physical properties.

Structural realism, as formulated by Ladyman [12] and elaborated by French and Ladyman [13], holds that the world's furniture consists not of objects with intrinsic, context-independent properties but of structural relations. Individual objects — particles, fields, spacetime points — are not the primary ontological constituents; relations and structures are. Phase Theory provides a concrete physical realization of this philosophical position: the admissible phase manifold M_{adm} is the fundamental structure, and all apparently object-like entities (particles, fields, spacetime) are emergent from the structural organization of this manifold.

Within Phase Theory, contextuality is the clearest manifestation of structural realism at the level of observable physics. The contextuality of outcome assignments — the dependence of what is determined on which context (structural partition) is imposed — is directly the statement that there are no intrinsic context-independent properties. The outcomes of context-specific determinations are structural relations between the phase configuration and the imposed partition, not revelations of pre-existing intrinsic properties. This is ontic structural realism made operationally concrete and experimentally testable.

Phase Theory also illuminates the debate between epistemic and ontic structural realism. Epistemic structural realism (ESR) [49] holds that our knowledge of science is knowledge of structural relations but does not

commit to the claim that structures are all that exist — intrinsic properties may exist but be unknowable. Ontic structural realism (OSR) holds that structures are ontologically primary and intrinsic properties are not merely unknown but absent. Phase Theory supports OSR: in Phase Theory, there are no intrinsic properties to be unknown. The phase configuration $c \in M_{\text{adm}}$ is structurally characterized — it is defined by its position in the relational structure of M_{adm} , not by any intrinsic properties. Context-relative outcomes are not epistemic limitations — they are the complete ontological story.

The apparently surprising nature of quantum contextuality — its resistance to classical intuition, its non-classical character — is, within Phase Theory, a consequence of taking classical noncontextual ontology as the default. Classical physics, in the contractible limit of M_{adm} , supports a structural ontology where global assignments exist and all partitions are compatible. This gives the appearance of intrinsic properties: outcomes are context-independent (because all contexts are compatible) and therefore appear to correspond to pre-existing values. Phase Theory shows this is an artifact of the classical limit: in the full theory, contextuality is the default, noncontextuality is the exception, and the appearance of intrinsic properties is the effective description valid only in the contractible regime.

Remark 33.2 (Underdetermination and Phase Theory)

Phase Theory provides a response to the standard underdetermination objection to scientific realism [50]. The underdetermination argument holds that any body of evidence is compatible with multiple incompatible theories, undermining realism. Phase Theory's falsifiable predictions (Section 34) are sufficiently specific and structurally grounded that they distinguish Phase Theory from all known alternatives — including standard quantum mechanics, Bohmian mechanics, and many-worlds. The falsifiability of Phase Theory is thus a direct argument for the possibility of scientific realism about structure: if the predictions are confirmed, we have evidence for the specific structural claim that reality is governed by global admissibility enforcement on a phase-bearing substrate.

The implications for the nature of physical properties extend beyond quantum mechanics. If contextuality is the default mode of structural reality — as Phase Theory asserts — then the concept of an intrinsic, context-independent physical property is a useful fiction valid only in the classical limit. All physical properties, in the full Phase Theory framework, are context-relative structural determinations. This has implications for the metaphysics of causation (causal relations are context-relative), identity (particle identity is context-relative), and persistence (persistence through time is context-relative persistence in M_{adm}). A full treatment of these implications is reserved for the philosophical companion to this series.

38. Conclusion

This paper has derived quantum contextuality as a necessary consequence of global admissibility enforcement on the phase-bearing substrate of Phase Theory. The derivation proceeds without invoking wavefunctions, hidden variables, measurement-induced collapse, or any ontological presuppositions imported from quantum mechanics. Contextuality emerges from the topology of the admissible phase manifold M_{adm} as a structural feature of phase-consistent reality.

The principal results are the following. First, we established that contextuality is equivalent to the non-existence of a global section of the outcome sheaf over the poset of admissible contexts — a purely structural characterization making no reference to agents or experiments (Definition 33.5, Proposition 33.3). Second, we showed that this non-existence is equivalent to the non-triviality of the first Čech cohomology class $[\sigma] \in H^1(\text{Ctx}(M_{\text{adm}}); F)$ (Theorem 33.17). Third, we derived all standard contextuality results — the Kochen–Specker theorem (Proposition 33.4), state-independent contextuality (Theorem 33.7), the Peres–Mermin magic

square (Theorem 33.13), graph-theoretic bounds (Theorem 33.14), and Bell nonlocality (Theorems 33.5–33.6) — as corollaries of Phase Theory topology.

Fourth, we developed contextuality monotones (Section 20) and showed their ordering $CF \leq R \leq G$ (Theorem 33.16). Fifth, we derived the Born rule as the contextual probability measure (Theorem 33.8) and the exclusivity principle as an admissibility theorem (Theorem 33.25). Sixth, we established the thermodynamic cost of contextuality (Theorem 33.26) and showed it implies the second law and the Landauer bound. Seventh, we derived the Gottesman-Knill theorem and the computational role of magic states from Phase Theory (Theorem 33.12, Corollary 33.6). Eighth, we resolved the measurement problem (Theorem 33.30) and provided structural derivations of Lorentz covariance (Theorem 33.22), gauge-contextuality correspondence (Theorem 33.28), and many-body contextuality scaling (Theorem 33.27).

Nine classes of falsifiable predictions beyond standard quantum mechanics were enumerated in Section 34, including: contextuality reduction at symmetry-breaking transitions, dark-sector contextuality imprints in recoil spectra, exact linear scaling of contextuality with entanglement entropy, and quantized contextuality monotones in topological phases. Each of these predictions distinguishes Phase Theory from standard quantum mechanics and from all other foundational approaches reviewed in Section 35.

Contextuality is not a quantum oddity. It is the structural signature of a phase-consistent universe. Reality does not contradict itself — but any description that ignores global admissibility structure will encounter apparent contradictions. Every context must be respected, every partition must be globally admissible, and every outcome must be understood as a context-relative structural selection, not the revelation of a pre-existing context-independent fact.

The recognition that contextuality is structural, not operational — that it belongs to the furniture of reality, not to the experimental choices of observers — resolves simultaneously the measurement problem, the mystery of non-commutativity, the inexplicability of the Tsirelson bound, the puzzle of PR box impossibility, and the apparent strangeness of quantum mechanics. All of these are manifestations of a single fact: the admissible phase manifold M_{adm} has non-trivial topology, and non-trivial topology obstructs global sections of outcome sheaves. This is contextuality — and it is, in Phase Theory, as inevitable as the curvature of a sphere.

Outlook. Future papers in the Phase Theory series will extend this framework to entanglement (Paper 34: "Entanglement as Topological Coupling on the Admissible Phase Manifold"), decoherence (Paper 35: "Decoherence as Admissibility Restriction: Classical Emergence from Phase Topology"), and the quantum-classical transition (Paper 36: "The Quantum-Classical Transition as Topological Simplification of M_{adm} "). The present paper provides the contextuality framework that underlies all three. Papers 37 and 38 will address Phase Theory cosmology and the dark energy problem, connecting the contextuality bounds of Section 34 to observable cosmological signatures.

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This paper is dedicated to the proposition that physics should explain, not merely describe — and that explanation, in the deepest sense, requires ontology.

DECLARATION:

This is a preprint under review. Results have not yet been independently verified.

The Phase Theory framework is the sole intellectual product of the author. January 2026. Phase Theory Research Program.

References

- [1] Kochen, S. and Specker, E.P. (1967). The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics* 17, 59-87.
- [2] Bell, J.S. (1964). On the Einstein-Podolsky-Rosen paradox. *Physics* 1, 195-200.
- [3] Clauser, J.F., Horne, M.A., Shimony, A. and Holt, R.A. (1969). Proposed experiment to test local hidden-variable theories. *Physical Review Letters* 23, 880-884.
- [4] Peres, A. (1990). Incompatible results of quantum measurements. *Physics Letters A* 151, 107-108.
- [5] Mermin, N.D. (1990). Simple unified form for the major no-hidden-variables theorems. *Physical Review Letters* 65, 3373-3376.
- [6] Spekkens, R.W. (2005). Contextuality for preparations, transformations, and unsharp measurements. *Physical Review A* 71, 052108.
- [7] Abramsky, S. and Brandenburger, A. (2011). The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics* 13, 113036.

- [8] Cabello, A., Severini, S. and Winter, A. (2014). Graph-theoretic approach to quantum correlations. *Physical Review Letters* 112, 040401.
- [9] Howard, M., Wallman, J., Veitch, V. and Emerson, J. (2014). Contextuality supplies the magic for quantum computation. *Nature* 510, 351–355.
- [10] Yu, S. and Oh, C.H. (2012). State-independent proof of Kochen–Specker theorem with 13 rays. *Physical Review Letters* 108, 030402.
- [11] Hanks, M. (2026). Phase Theory: A unified theory of matter, light, and geometry. Phase Theory Research Program Preprint Series, Papers 1–32.
- [12] Ladyman, J. (1998). What is structural realism? *Studies in History and Philosophy of Science* 29, 409–424.
- [13] French, S. and Ladyman, J. (2003). Remodelling structural realism: Quantum physics and the metaphysics of structure. *Synthese* 136, 31–56.
- [14] Tsirelson, B.S. (1980). Quantum generalizations of Bell's inequality. *Letters in Mathematical Physics* 4, 93–100.
- [15] Popescu, S. and Rohrlich, D. (1994). Quantum nonlocality as an axiom. *Foundations of Physics* 24, 379–385.
- [16] Gachechiladze, M., Budroni, C. and Gühne, O. (2016). Extreme violation of local realism in quantum hypergraph states. *Physical Review Letters* 116, 070401.
- [17] Conway, J. and Kochen, S. (2006). The free will theorem. *Foundations of Physics* 36, 1441–1473.
- [18] Cabello, A. (2013). Simple explanation of the quantum violation of a fundamental inequality. *Physical Review Letters* 110, 060402.
- [19] Abramsky, S., Mansfield, S. and Barbosa, R.S. (2017). The cohomology of non-locality and contextuality. *Electronic Notes in Theoretical Computer Science* 298, 1–14.
- [20] Hardy, L. (1993). Nonlocality for two particles without inequalities for almost all entangled states. *Physical Review Letters* 71, 1665–1668.
- [21] Bohr, N. (1935). Can quantum-mechanical description of physical reality be considered complete? *Physical Review* 48, 696–702.
- [22] Bohm, D. (1952). A suggested interpretation of the quantum theory in terms of "hidden" variables. *Physical Review* 85, 166–179.

- [23] Mermin, N.D. (1993). Hidden variables and the two theorems of John Bell. *Reviews of Modern Physics* 65, 803–815.
- [24] Einstein, A., Podolsky, B. and Rosen, N. (1935). Can quantum-mechanical description of physical reality be considered complete? *Physical Review* 47, 777–780.
- [25] von Neumann, J. (1932). *Mathematische Grundlagen der Quantenmechanik*. Springer, Berlin. [English translation: Princeton University Press, 1955.]
- [26] Cabello, A. (2010). Proposal for revealing quantum nonlocality via local contextuality. *Physical Review Letters* 104, 220401.
- [27] Klyachko, A.A., Can, M.A., Binicioğlu, S. and Shumovsky, A.S. (2008). Simple test for hidden variables in spin-1 systems. *Physical Review Letters* 101, 020403.
- [28] Fine, A. (1982). Hidden variables, joint probability, and the Bell inequalities. *Physical Review Letters* 48, 291–295.
- [29] Kirchmair, G., Zähringer, F., Gerritsma, R., Kleinmann, M., Gühne, O., Cabello, A., Blatt, R. and Roos, C.F. (2009). State-independent experimental test of quantum contextuality. *Nature* 460, 494–497.
- [30] Hensen, B. et al. (2015). Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. *Nature* 526, 682–686.
- [31] Giustina, M. et al. (2015). Significant-loophole-free test of Bell's theorem with entangled photons. *Physical Review Letters* 115, 250401.
- [32] Abramsky, S., Barbosa, R.S. and Mansfield, S. (2017). Contextual fraction as a measure of contextuality. *Physical Review Letters* 119, 050504.
- [33] Gottesman, D. (1998). Theory of fault-tolerant quantum computation. *Physical Review A* 57, 127–137.
- [34] Bravyi, S. and Kitaev, A. (2005). Universal quantum computation with ideal Clifford gates and noisy ancillas. *Physical Review A* 71, 022316.
- [35] Hein, M., Eisert, J. and Briegel, H.J. (2004). Multiparty entanglement in graph states. *Physical Review A* 69, 062311.
- [36] Vidal, G. (2000). Entanglement monotones. *Journal of Modern Optics* 47, 355–376.

- [37] Raussendorf, R. (2013). Contextuality in measurement-based quantum computation. *Physical Review A* 88, 022322.
- [38] Hartshorne, R. (1977). *Algebraic Geometry*. Springer Graduate Texts in Mathematics, Vol. 52. Springer, New York.
- [39] Bennett, C.H. and Wiesner, S.J. (1992). Communication via one- and two-particle operators on Einstein-Podolsky-Rosen states. *Physical Review Letters* 69, 2881–2884.
- [40] Bennett, C.H., Brassard, G., Crépeau, C., Jozsa, R., Peres, A. and Wootters, W.K. (1993). Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels. *Physical Review Letters* 70, 1895–1899.
- [41] Sorkin, R.D. (1994). Quantum mechanics as quantum measure theory. *Modern Physics Letters A* 9, 3119–3127.
- [42] Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development* 5, 183–191.
- [43] Everett, H. (1957). "Relative state" formulation of quantum mechanics. *Reviews of Modern Physics* 29, 454–462.
- [44] Wallace, D. (2012). *The Emergent Multiverse: Quantum Theory according to the Everett Interpretation*. Oxford University Press, Oxford.
- [45] Rovelli, C. (1996). Relational quantum mechanics. *International Journal of Theoretical Physics* 35, 1637–1678.
- [46] Fuchs, C.A., Mermin, N.D. and Schack, R. (2014). An introduction to QBism with an application to the locality of quantum mechanics. *American Journal of Physics* 82, 749–754.
- [47] Bombelli, L., Lee, J., Meyer, D. and Sorkin, R.D. (1987). Space-time as a causal set. *Physical Review Letters* 59, 521–524.
- [48] Maudlin, T. (1995). Three measurement problems. *Topoi* 14, 7–15.
- [49] Worrall, J. (1989). Structural realism: The best of both worlds? *Dialectica* 43, 99–124.
- [50] Stanford, K. (2017). Underdetermination of scientific theory. In *Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University.

- [51] Lovász, L. (1979). On the Shannon capacity of a graph. *IEEE Transactions on Information Theory* 25, 1-7.
- [52] Veitch, V., Mousavian, S.A.H., Gottesman, D. and Emerson, J. (2014). The resource theory of stabilizer computation. *New Journal of Physics* 16, 013009.
- [53] Atiyah, M.F. and Singer, I.M. (1963). The index of elliptic operators on compact manifolds. *Bulletin of the American Mathematical Society* 69, 422-433.
- [54] Kitaev, A. (2003). Fault-tolerant quantum computation by anyons. *Annals of Physics* 303, 2-30.
- [55] Wen, X.G. (2004). *Quantum Field Theory of Many-Body Systems*. Oxford University Press, Oxford.
- [56] Hatcher, A. (2002). *Algebraic Topology*. Cambridge University Press, Cambridge. [Freely available at math.cornell.edu/~hatcher.]
- [57] MacLane, S. and Moerdijk, I. (1992). *Sheaves in Geometry and Logic: A First Introduction to Topos Theory*. Springer, New York.
- [58] Milnor, J. and Stasheff, J.D. (1974). *Characteristic Classes*. Annals of Mathematics Studies, Vol. 76. Princeton University Press, Princeton.
- [59] Budroni, C., Cabello, A., Gühne, O., Kleinmann, M. and Larsson, J.-Å. (2021). Kochen-Specker contextuality. *Reviews of Modern Physics* 94, 045007.
- [60] Zukowski, M. and Brukner, Č. (2012). Bell's theorem for general N-qubit states. *Physical Review Letters* 88, 210401.

¹ The Phase Theory series currently comprises 33 papers. Papers 1-10 establish the phase ontology and emergent spacetime geometry. Papers 11-20 derive the particle spectrum and interaction structure. Papers 21-32 develop quantum mechanics, thermodynamics, and cosmology from Phase Theory. The present paper (Paper 33) addresses contextuality.

² Throughout this paper, "admissible" means consistent with the global phase consistency functional $\Phi[c] = 0$. Inadmissible configurations are physically unrealized in Phase Theory.

³ The notation M_{adm} is used consistently throughout this series for the admissible phase manifold. Its precise mathematical definition depends on the physical system under consideration; the general properties established here hold in full generality.

⁴ Equation numbering follows the convention (section.equation) within sections. Theorems, definitions, propositions, and corollaries are numbered 33.X throughout, consistent with the paper number in the Phase Theory series.

⁵ The connection to the Mochizuki-Yamashita conjecture in arithmetic geometry, relating the structure of étale fundamental groups to Faltings' theorem, is noted as a potential direction for future mathematical development of the Phase Theory partition lattice formalism.